

Intelligent assessment of power quality disturbances: A comprehensive review on machine learning and deep learning solutions

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ARTICLE INFO

Keywords:

Power quality assessment
Artificial intelligence
Deep learning
Machine learning
Digital signal processing
Feature extraction

ABSTRACT

As the global focus on clean energy and smart grids intensifies, detecting power quality disturbances (PQDs), caused by energy instability, has become increasingly critical for achieving sustainable development goals by ensuring stable and reliable power quality. Power quality disturbances, such as voltage sags or harmonics, are disruptions in the electrical supply that can affect everything from household appliances to industrial machinery, making their detection and management essential for a stable power system. They can cause significant damage to power grid infrastructure, leading to energy inefficiency, restricted electricity generation and consumption, equipment malfunction, and industrial process failures. The incorporation of artificial intelligence (AI) has transformed PQD classification, providing substantial advancements in monitoring and managing electrical systems. This paper presents a systematic review of the existing literature, focusing on the integration of machine learning and deep learning techniques for PQD detection. It analyzes high-quality studies on PQDs detection and classification, categorizing them based on the AI techniques employed. Additionally, it emphasizes the role of digital signal processing (DSP) techniques in extracting features, with studies segregated based on the incorporation of DSP and non-DSP approaches. A case study demonstrates the practical application and effectiveness of AI techniques in real-world contexts, with the Bagged Trees classifier achieving the highest testing accuracy of 96.6 %. The insights provided aim to support researchers and practitioners in navigating the evolving landscape of power quality assessment, ultimately improving the reliability and accuracy of PQD detection systems.

1. Introduction

In early times, power quality problems were used to define the circumstances in which the distribution of power was affected. However, nowadays, after significant technological advancements, the definition of power quality has evolved, it is explicitly defined as the set of certain features of electricity at a particular location within an electrical system. Power quality is now the assessment of

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parameters such as voltage, current, and frequency to ensure they remain within specified limits [1]. It ensures that good quality power is maintained at all the levels of electricity networks that is generation, transmission, and distribution levels [2–4]. Poor power quality causes trouble in transmission and electronic equipment malfunctions like adjustable speed drivers (ASD's), data network congestion, flickering screens, unwanted activation of protective devices, and many more, thereby, reducing the lifespan of equipment [5–6]. With the decentralization of electricity suppliers, power quality (PQ) has become a key differentiating factor when choosing between two electricity companies for a consumer [7–9].

Any disturbance in the power grid is known as Power Quality Disturbance or PQD. The growing use of solid-state switching devices, lighting controls, rectifiers, inverters, data processing equipment, non-linear loads, and relaying equipment has contributed to PQ disturbances. Deviations from acceptable limits can lead to various issues like voltage sag, flicker, voltage swell, transient events, and harmonics, all of which negatively impact power quality. These disturbances can be categorized into three main types: magnitude variations, steady-state harmonics, and sudden transients. The magnitude variations in voltage or current such as interruption, swell, and sag have been observed in real-time power system due to the PQ culprits such as the inception of line faults and fluctuations in load conditions. Similarly, impulsive and oscillatory transients are generated due to abrupt changes in the operating conditions e.g., switching of capacitor banks, lightning, etc. The steady-state PQ disturbances namely harmonics, notches, and flickers typically occur due to the use of power electronic converters and non-linear loads [10]. These PQ disturbances can lead to significant costs associated with the malfunction of equipment. Therefore, effective PQ monitoring is a need of the day to restore the normal power supply after mitigating the disturbances. A few years ago, the primary concern for electricity consumers was the reliability of supply which means continuity of electricity service. But nowadays, PQ monitoring has become a crucial subject for the modern power industry, aimed at safeguarding electronic and electrical equipment and identifying the causes of disturbances [11]. Certain sensitive loads, such as air traffic control systems, hospitals (life support, operation theatres, patient database systems), processing plants (fabric, food, semiconductor, rayon, etc.), and bank security systems, require clean and uninterrupted power. According to the European PQ survey report, PQ issues result in a financial loss of approximately 150 billion Euros annually in the EU-25 countries [12]. Due to these stringent requirements, the detection of even a small power quality disturbance becomes critically important. The assessment of power quality increases uptime and helps in the prevention of equipment failures.

Power Quality standards, as mentioned in Fig. 1, are set to define the electromagnetic phenomenon related to PQD, helping industrial and commercial entities recognize deviation in normal voltage or current conditions. These are developed to ensure the safety, reliability, and performance of electrical products and systems. International Electrotechnical Commission (IEC) standards consist of a series of standards known as Electromagnetic Compatibility (EMC) standards. It has six parts, namely IEC 61,000–1-X (general), IEC 61,000–3-X (limits), IEC 61,000–2-X (environment), IEC 61,000–4-X (testing and measurement techniques), IEC 61,000–6-X (generic standards) and IEC 61,000–5-X (installation and mitigation guidelines) [13–14]. Institute of Electrical and Electronics Engineers (IEEE) Standards are based on the theoretical background of power quality these include IEEE standard 1100 (supply for electronic), IEEE standard 519 (harmonic control), IEEE standard 1159 (monitoring), IEEE standard 1366 (reliability indices) [15,16–18]. Under European Norms (EN), EN 50,160 standard defined the acceptable deviation threshold of root mean square value (RMS) of voltage and current from their nominal levels during PQD [19]. Power Quality Assessment offers numerous advantages such as improvement in the

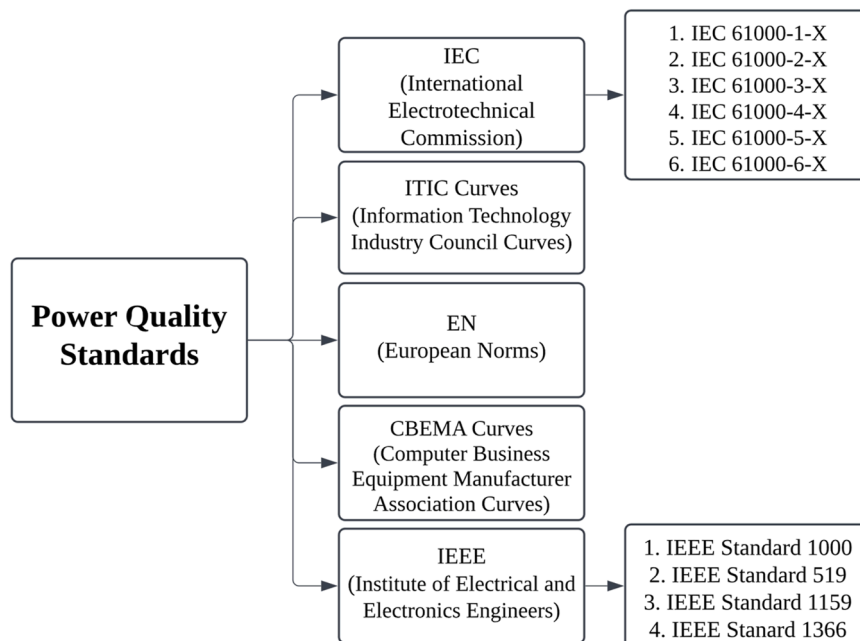


Fig. 1. Power Quality Standards Classification Chart.

quality and continuity of power in the plant, identification of the root cause of electrical failures and the areas for energy savings on the plant floor, and reduce the maintenance and production costs associated with electrical downtime.

According to the 2023 McKinsey rep ort, 55 % of organizations now use Artificial Intelligence in at least one business unit or function, up from 50 % in 2022 and 20 % in 2017. AI is increasingly utilized in various facets of the electrical field such as the diagnosis of a system, fault diagnosis, load forecasting, and many more. It is transmogrifying the management and operation of power systems. A broad subclassification of AI is depicted in Fig. 2. Machine Learning, a subset of AI, involves systems learning from data, recognizing patterns, and making decisions. Deep Learning, a further subset of Machine Learning, is inspired by the human brain's decision-making abilities and leverages multilayered neural networks. Especially in the realm of power quality disturbance detection, AI performs a pivotal role due to its pioneering abilities in data analysis, pattern recognition, and real-time monitoring. The conventional methods are oftentimes inaccurate and take human effort and time. Here, AI techniques notch up due to their supremacy in processing huge amounts of data and identification of intricate patterns. They enrich the overall system reliability, predictive maintenance, scalability, and efficiency.

The approaches employed for identifying and classifying power system issues typically involve multiple stages: collection of data, pre-processing, extraction of features, and classification as shown in Fig. 2. Some researchers also incorporate a feature selection stage after feature extraction and before feeding the data into the classification algorithm, known as the optimization stage. Feature extraction from signals can be performed using direct methods, such as calculating the RMS value of raw samples, or through digital signal processing (DSP) techniques like the Fourier transform, Kalman filter, wavelet transform (WT), S-Transform (ST), Hilbert-Huang Transform, and correlation feature selection. The features extracted can then be directly fed into classification algorithms. The extracted features are further applied to classifier algorithms, which automatically perform the classification of the data.

Sometimes, before this classification stage, the data undergoes an optimization process to enhance the feature and thus result in better accuracy and efficiency of the classification algorithm. Techniques used for optimization may include genetic algorithms, particle swarm optimization, or other heuristic methods. Classifier algorithms, including artificial neural networks, wavelet networks, support vector machines, fuzzy-expert systems, and rule-based systems as well as other sophisticated machine learning models, are prevalently employed in the classification of power quality (PQ) disturbances. The aforementioned methodology represents a conventional approach.

Nonetheless, research in this domain predominantly bifurcates into two distinct categories. Certain research employs DSP techniques for feature extraction, meticulously refining the data before inputting it into classification algorithms, while other researchers bypass this intermediary step, opting to directly feed the raw data into the classification models. Based on this, the research can be classified into two categories. Table 1 summarizes the contributions of a few studies in the field, detailing ML or DL technique applied, along with the employed DSP technique. It has been extended further, with more details, in the sections 4 and 5.

By systematically reviewing and comparing the methodologies, the paper provides valuable insights into their respective strengths and weaknesses, offering a thorough understanding of the current landscape in this field. The review focuses on the transformative impact of AI techniques, such as machine learning (ML) and deep learning (DL), in enhancing the accuracy and efficiency of PQD detection and classification. While PQD classification has garnered significant attention, a notable gap persists in existing reviews—namely, the lack of a unified framework that categorizes studies based on both AI methods and the use of digital signal processing (DSP) and non-DSP techniques. Both DSP and non-DSP techniques are a double-edged sword, offering distinct advantages and challenges in feature extraction. Recognizing this gap, this paper organizes the review by categorizing studies based on both the AI methodology employed and the use of DSP and non-DSP techniques. This dual perspective reveals emerging trends, identifies key areas of improvement, and bridges the divide between these diverse approaches. The purpose is to provide a structured analysis that not only highlights the advancements in AI-driven PQD detection but also underscores the role of DSP techniques in enhancing classification performance. Additionally, a case study is included to validate the findings and demonstrate the practical application of these methodologies.

The major contributions of this paper include a comprehensive categorization of research, not only based on the AI techniques employed but also emphasizing the role of DSP techniques in feature extraction. The review delves into a selection of high-quality research papers that focus on the detection and classification of various PQDs such as sag, swell, harmonics, and more. In addition to this exploration, it offers a comparative analysis of these works, shedding light on their methodologies, advantages, and limitations, enabling a clearer understanding of the state-of-the-art in PQD detection. Detailed explanations and algorithms of each AI technique are thoroughly presented before reviewing their specific applications within the domain. This approach allows for a deeper understanding of the underlying mechanisms that drive successful PQD detection and classification. To demonstrate the practical relevance and effectiveness of AI techniques in real-world scenarios, a case study is also included. By offering a thorough analysis of current trends, challenges, and innovations, the paper serves as a valuable guide for advancing research and practical applications in the field.

The paper has been structured in the following manner. Section 2 provides an overview of Power Quality Disturbances (PQDs) while Section 3 provides an assessment of PQDs, discussing various approaches for identifying and measuring PQDs. Section 4 explores the automatic classification of PQDs using machine learning techniques and Section 5 deals with deep learning techniques. Both provide a comprehensive analysis of various ML and DL methods along with the segregation of studies incorporating DSP techniques or not. Section 6 presents a case study to illustrate the practical application and effectiveness of the discussed classification techniques in real-world settings. Finally, Section 7 concludes the paper by summarizing the findings and outlining potential areas for future research.

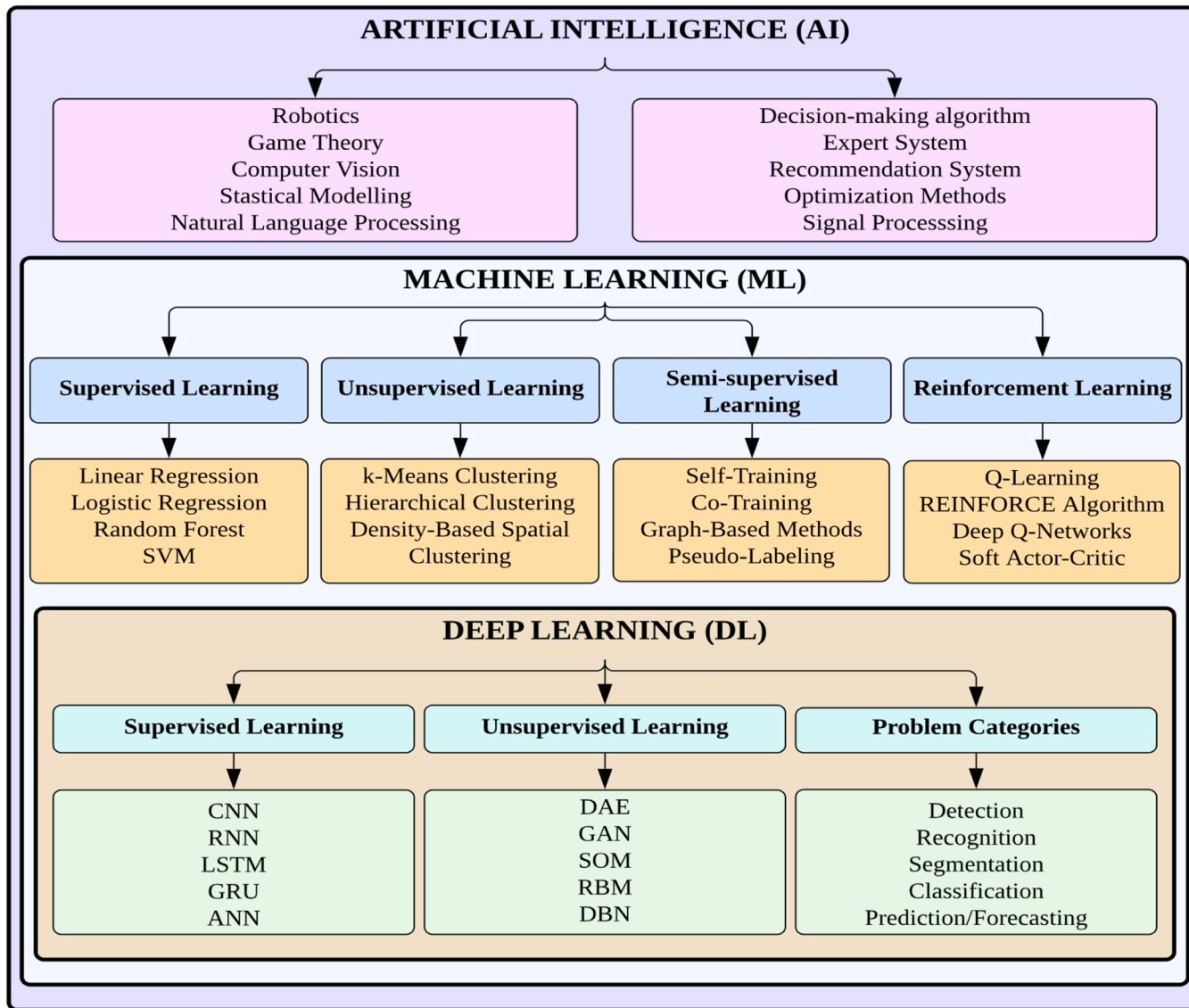


Fig. 2. Hierarchical Breakdown of Artificial Intelligence Algorithms and Applications (including Deep Learning and Machine Learning).

Table 1
Outline of key contributions to PQD analysis.

Reference	Contributions	DSP Technique	ML Technique	DL Technique
[29]	The study evaluates the performance of SVM with one-against-all (OAA) and other classification methods.	Hilbert-Huang Transform	Support Vector Machine	×
[31]	An effective SVM-based method using phase space reconstruction for accurate identification and classification of diverse power system disturbances.	×	Support Vector Machine	×
[33]	Develops a VMD-based feature extraction method for classifying power quality disturbances, achieving high accuracy with the Random Kitchen Sink (RKS) algorithm over SVM.	×	Support Vector Machine and Random Kitchen Sink	×
[36]	The Fast Discrete S-Transform (FDST) variants reduce the computational cost and maintain high accuracy.	Fast S-Transform	Decision Tree	×
[40]	The proposed PQ detection framework using a modified Stockwell transform and random forests achieve high accuracy with resilience to noise.	Modified S-Transform	Random Forest	×
[50]	The approach used allows for easier implementation and faster processing due to reduced and simple set of features.	S-Transform	Rule based classifier	×
[52]	A novel PQ event detection and classification method using 3-D space representation, PCA-based feature extraction, and neuro-fuzzy classifier.	Wavelet Transform	Neuro-fuzzy	×
[60]	The overall system's efficiency enhanced by the RBF network which requires less training time compared to other ANN architectures.	Hilbert Transform	×	Artificial Neural Network
[69]	The system's accuracy increased due to the combination of SSA and FDCT methods with DCNN.	Curvelet Transform (CT)	×	Deep Convolutional Neural Networks (DCNNs) and multiclass SVM
[74]	The hierarchical extreme learning machine (H-ELM) classifier accelerates the training process and provides robust classification.	Discrete Wavelet Transform	×	Hierarchical extreme learning machine (H-ELM)

2. Power quality disturbances

The capability of the power systems to deliver undistorted voltage, current, and frequency signals, is known as the quality of power supply. The disturbances can originate in a power supply at either the point of generation transmission or distribution level. Each PQD is attributed to a unique cause and result. As per IEC's standard electromagnetic series IEC-61,000, some of the phenomena that cause electromagnetic disturbances are conducted low-frequency, radiated low-frequency, conducted high-frequency, radiated high-frequency, electrostatic discharge (ESD), and nuclear electromagnetic pulse (NEMP). Fig. 3 presents a comprehensive classification flowchart of PQDs, incorporating various standards, including European, IEEE, and IEC standards, along with their detailed subclassifications.

According to EN standards, they can be classified into two categories that is, power quality events and power quality variations [19]. The power quality events encompass a range of disturbances that can disrupt the stability and reliability of electrical systems. Interruption happens when load current or supply voltage remains 1–10 % of the nominal value, classified as long (over 3 min) or short (under 3 min). The power quality events involve voltage swell, interruption, sag, and transients. Voltage sag involves a 10–90 % reduction of the nominal voltage, quickly recovering within a few cycles to seconds. Conversely, voltage swells entail a temporary increase of 10–90 % in the nominal voltage, also followed by a rapid recovery. Transients are short-lived deviations in current or voltage waveforms, lasting less than half a cycle.

The power quality variations cover voltage frequency variation, voltage magnitude variations, voltage unbalance, voltage fluctuation flicker, and waveform distortion. The voltage frequency variations occur due to the imbalance in the fluctuations in power levels between generation and consumption. The voltage magnitude variation results from fluctuation in power flow, manifesting as undervoltage or overvoltage, originating in changes in active power and reactive power, or even due to small load fluctuations. Voltage unbalance occurs whenever there is a disparity in the phase voltages (RMS value) or phase angles across consecutive phases in a three-phase system. Voltage fluctuations arise from varying load currents and unpredictable changes in voltage envelope. Waveform distortion refers to the non-sinusoidal waveforms of voltage and current, primarily originating from non-linear elements within the power system network, which generate non-sinusoidal currents. Table 2 provides a comprehensive overview of power quality variations and events, detailing their respective origins and associated effects.

3. Assessment methodology of power quality disturbances

A Power Quality Disturbance is an electromagnetic phenomenon which can not only damage the equipment but also has the capability to destroy the complete process. This can lead to significant economic losses. Assessment of PQDs helps in identifying and mitigating the causes and ensuring a reliable and stable power system. Traditionally power quality disturbances have been detected through visual inspection, which has been a challenging task for PQ engineers due to the large volume of data involved in power quality waveform monitoring. This manual process is inflexible and difficult to manage, making the development of automatic devices for PQD

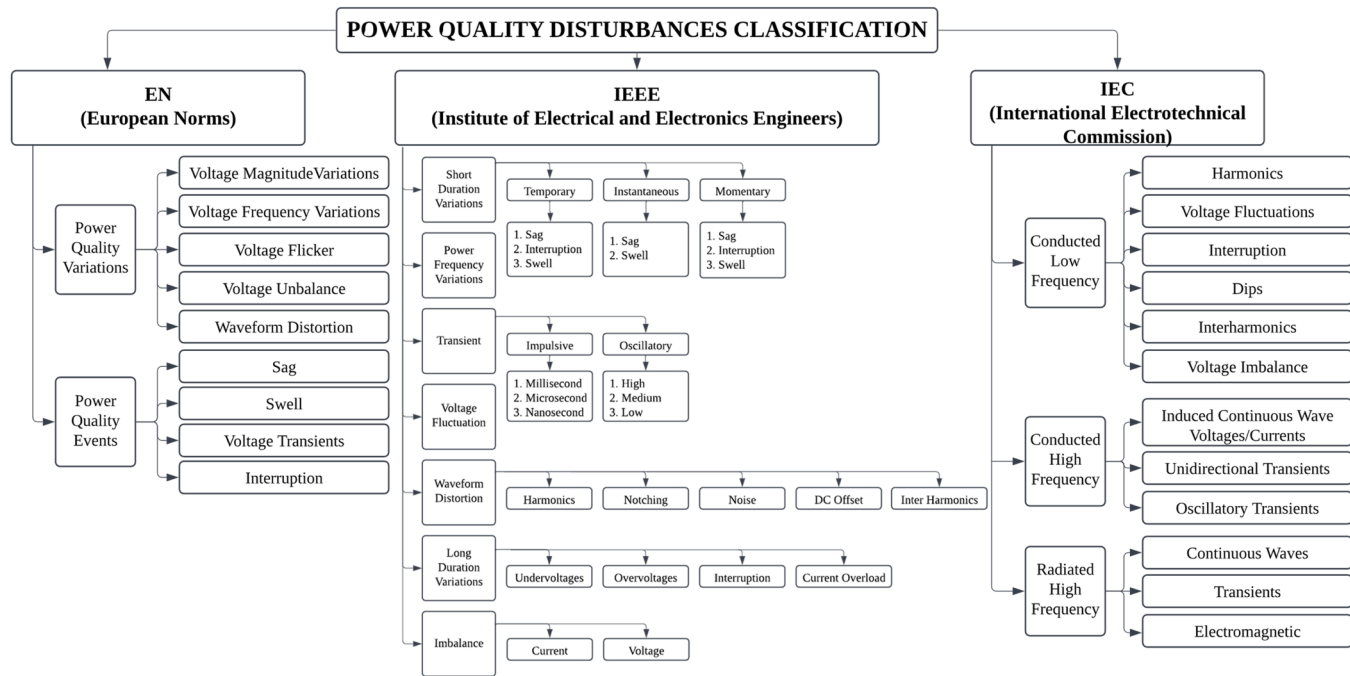


Fig. 3. Classification of PQDs based on Different Standards.

Table 2
Overview of PQDs in Alignment With EN Standards.

Power Quality Disturbances	Origin	Effects
Power Quality Variations		
Voltage Frequency Variations	Power imbalance between generation and consumption	- Minimal impact on end-user equipment - Minor time variations in clocks and slight speed variations in induction/synchronous motors
Voltage Magnitude Variations	- Active and reactive power flow variations - Small load variations	May reduce equipment lifetime (e.g., incandescent lamps)
Voltage Unbalance	- Large amount of transportation of power over long distances - Single/two-phase faults until circuit breaker trips	- Unbalanced currents in three-phase rectifiers and rotating machines - Results in heating or losses in cables and lines
Voltage Fluctuations and Flicker	- Load current fluctuations - Light flickers are due to fluctuations in the frequency range of 1 to 20 Hz.	- Little effect on non-incandescent lamps - Affects control systems and motor acceleration/braking
Waveform Distortion	- Non-linear power system elements - Industrial loads (inverters, rectifiers) - Electric arc loads (welding machines, lighting) - Domestic loads (ovens, televisions)	- Circuit breaker tripping - transformer heating - Overheating of neutral conductor - Time variations in digital clocks - Malfunctioning of equipment
Power Quality Events		
Interruptions	- During fault removal by protection devices - Incorrect or intentional interrupting device opening - Equipment failure	- Unnecessary protective device tripping - Equipment malfunction - Halt customer facility - Voltage frequency reduction Information loss
Voltage Sag	- Increase in current for short duration - Starting of motor - Transformer energizing - Line faults (symmetrical and unsymmetrical)	- Equipment malfunction - Relay and contactor tripping Data loss in computers Speed reduction in asynchronous motors
Voltage Swell	- Short-duration voltage swell due to single-phase faults - Long-duration voltage swell due to reactor switching for power factor control	- Screen/light flickering - Data loss
Transients	- Capacitor/inductor energizing or de-energizing - Self-clearing faults - Lighting strokes	- Stoppage of sensitive equipment - Unusual equipment damage - Misoperation or lockup of computer and electronic equipment

measurement highly desirable. Recently, the integration of signal processing techniques with artificial intelligence has significantly contributed to the automatic classification of PQD waveforms [20]. Most of the available datasets for PQD classification techniques are non-standard, varying in disturbance type, sample sizes, and data labels. Thus, the creation of synthetic data become increasingly important for good analysis. Synthetic data also allows the simulation of a wide range of distance scenarios that may not be easily captured in real-world conditions. It also facilitates the training of machine learning models, providing them with a diverse and comprehensive dataset [21].

The assessment procedure involves many key steps to identify, analyze and address PQDs, as detailed in Fig. 4. From collection of data to identification and classification of disturbances, measurement of severity of disturbances, and analysis of its root cause. The collection of data, a pivotal step, involves recording voltage, current, and frequency data through power quality instruments or meters including PQ monitors. These monitors play a crucial role in transforming raw data into useful information. They are currently helping modern industries to identify the root causes of PQDs. After this, the preprocessing and feature selection stages refine the data collected and extract the most informative features. Numerous studies have employed various Digital Signal Processing techniques for feature extraction, including the Fourier transform, Kalman filter, Wavelet transform, S-transform, Hilbert-Huang transform, and Gabor transform. Other commonly used Fourier transforms include the Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT), and Short-Time Fourier Transform (STFT). Methods like Gabor transform, Chirp transform, and Slant transform often face challenges

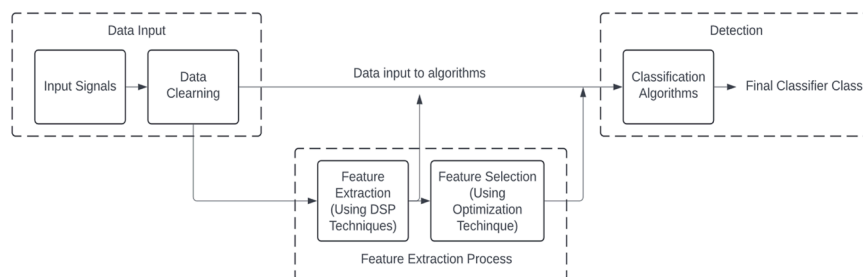


Fig. 4. Standardized Methodology for PQD Classification.

related to low-frequency resolution. To address this limitation, model-based approaches such as Multiple Signal Classification (MUSIC) and Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) are employed. AI techniques used for the purpose of optimization include the selection of optimal features for accurate classification. These chosen features are directly fed into the learning algorithm. Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Bee Colony Optimization (BCO) are some of the optimization techniques.

Approaching the last and critical step, PQD identification and classification. Several AI techniques commonly known as classifiers are used for the categorization of the type of PQ disturbance. These techniques can also be subdivided into two categories- supervised and unsupervised techniques. Regression technique (where the output variable is continuous) and Classification technique (where the output variable is categorical) are the two major divisions of supervised learning. The major classifiers include Linear machines, Artificial Neural Networks (ANN), Bayesian classifiers, Kernel machines, Support Vector Machines (SVM), Rule-based expert systems, and fuzzy classification systems. Lastly, the severity of disturbances can be measured through indices like Voltage Distortion, Flicker Severity Index, and Total Harmonic Distortion. An alternative approach is the incorporation of non-DSP methodologies. This approach bypasses the conventional reliance on signal transformation, opting instead for techniques that process raw data directly. This often includes ML algorithms that extract features from raw data, as well as DL models like CNNs and Transformers that seamlessly integrate feature extraction and classification. By eliminating the dependence on DSP transformations, non-DSP approaches provide a versatile and scalable alternative.

4. Automatic classification OF PQDS using machine learning technique

AI plays a pivotal role in automating the classification of PQDs, greatly enhancing, the accuracy and efficacy of detection. The AI techniques excel in processing large volumes of data, adopting novel types of disturbances, and continuously improving through learning. The automatic classification of PQD becomes essential due to the increasing complexity and frequency of power disturbances in modern electrical grids. Manual classification is not only labor intensive but also prone to errors, making AI-driven solutions dispensable for real-time monitoring and effective management of PQ.

An optional optimization stage, occurring after feature extraction and before classification, is also adopted. Feature extraction can be achieved either through direct metric calculations, such as determining the RMS value of raw data or by employing transformation-based techniques. Before classification, data may undergo optimization using methods like genetic algorithms, colony optimization, or other heuristic techniques. The extracted features are then fed into a classification algorithm, which can be an artificial neural network or machine learning technique such as fuzzy expert systems, support vector machine (SVM), random forest (RF), etc. These algorithms automatically classify the data based on the optimized feature vectors, ensuing accuracy and efficient PQD detection. The research in this domain is divided into two main approaches. One approach utilizes DSP techniques for feature extraction, extracting the features before the classification stage. The other approach skips this step and feeds raw data directly into a classification algorithm. The process involved in separating the studies into DSP-based and non-DSP-based approaches entails identifying the feature extraction techniques used in each study. The remaining paper delves into a comprehensive exploration of various AI techniques and their underline mathematical models. It highlights key studies that have employed these techniques for classification, comparing those that incorporate DSP techniques with those that have opted to include them. This thorough analysis aims to provide a nuanced understanding of the effectiveness and applicability of both approaches in the classification of PQDs.

4.1. Support vector machine

SVM initially proposed by Corinna Cortes and Vladimir Vapnik in 1995 [22], was originally termed “Support-Vector Networks”. This binary classification algorithm revolutionaries by mapping vectors non-linearly to a high dimensional feature space, thereby constructing a hyper plane (linear decision surface). This approach effectively handles both separable and non-separable. The hyperplane derived is optimal as it maximizes the margin between classes based on the training data [23]. SVM’s ability to achieve significant results makes it a pivotal tool in machine learning applications. SVMs excel at handling linearly separable problems, where the objective lies in identifying an optimal hyperplane that maximizes the separation between two classes. As in a two-dimensional space, multiple lines can separate the classes, but the optimal line maximizes the margin between the closest point of each class, known as support vectors (SVs). The optimal hyperplane is defined by the equation.

$$w^T + b = 0 \quad (1)$$

Where w is the weights vector and b is the threshold. The relation between x_i and $f(x_i)$ is:

$$f(x_i) = w^T x_i + b \quad (2)$$

As the optimal hyperplane is equivalent to the maximal distance between the closest vectors to the hyperplane, the Euclidean distance is defined between the nearest points and the hyper plane $f(x)$ as:

$$r = \frac{|f(x)|}{\|w\|} \quad (3)$$

Where $f(x)$ refers to the functional margin. Maximizing the norm of the weight vector.

$$\min \frac{1}{2} \|w\| \text{ subject to : } y_i(w^T x + b) \geq 1, i = 1, 2, \dots, n \quad (4)$$

The solution to this optimization problem can be obtained at the saddle point of the Lagrange function:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i(w^T x + b)] \quad (5)$$

where α_i is Lagrange multipliers, subject to $\alpha_i \geq 0$ and Eq. (6) has to be minimized with respect to w, b and maximized with respect to α_i . This leads to the dual problem. This optimization problem is solved using Lagrange multipliers, leading to the dual problem.

$$L(w, b, \alpha) = \sum_{i=1}^n \alpha_i + \frac{1}{2} \sum_{i=1, j=1}^n \alpha_i \alpha_j y_i y_j (x_i)^T x_j \quad (6)$$

Let us suppose $\bar{\alpha}_i$ is the result of Eq. (6), X_r is a support vector belonging to the positive class and X_s belongs to the negative class. Then the optimal solution w and b can be expressed as:

$$\bar{w} = \sum_{i=1}^n \bar{\alpha}_i y_i x_i \quad (7)$$

$$\bar{b} = -\frac{1}{2} [\bar{w}^T (X_r + X_s)] \quad (8)$$

The resultant final decision function can be expressed as follows:

$$f(x) = \text{sgn} \left[\sum_{SVs} \bar{\alpha}_i y_i (x_i)^T x + \bar{b} \right] \quad (9)$$

In the case of non-linearly separable problems, the decision function in higher dimensional space is

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \quad (10)$$

4.1.1. With the incorporation of DSP techniques

SVM classifiers have attracted considerable interest from power system researchers for their ability to generate optimal, sparse solutions by minimizing both generalization and training error while maximizing the margin of separation in high dimensional space [24]. The authors [25] introduced an innovative PQ event recognition system that employs a k-means-based Apriori algorithm for the feature section to efficiently extract distinctive features from PQ events. It also includes normalization and segmentation of three-phase event voltage signals, followed by a nine-level multiresolution analysis using wavelet transform to derive wavelet coefficients from the signals. By utilizing a Least Square SVM classifier, the system achieves a remarkable performance of 98.88 %. Peter G. V. Axelberg et al. [26] presented a novel SVM-based method for classifying common voltage disturbances, using individually trained SVMs as nodes in a binary decision tree for multiclass classification, demonstrating classification accuracy of 92.80 %. A robust method against noisy and noiseless signals is presented in the paper [24]. The method involves three stages: extraction of features using wavelet transform (WT) and other techniques, feature selection using sequential forward selection (SFS) to obtain the most accurate features, and determining the optimal input vector for the SVM classifier. The test accuracies obtained for pure signal and noisy signals with 30db and 20db noises are 99.43 %, 98.63 %, and 97.75 % respectively.

In [27], a novel approach has been introduced using wavelet transform and a self-organizing learning array (SOLAR) system for classifying PQDs. It demonstrated effective feature extraction via multi-resolution analysis and showcased SOLAR's advantages over traditional neural networks. The test accuracies for pure signals and noisy signals with 30db and 20db noises were obtained as 94.93 %, 92 %, and 93 % respectively. In [28], the wavelet packet transforms the entropy feature extraction technique and the weighted SVM is employed to classify various types of disturbances. The algorithm achieved a classification accuracy of 98.4 % for pure signals. The authors in [29] proposed an innovative PQ monitoring approach combining Hilbert-Huang transform (HHT) and Ensemble Empirical Mode Decomposition (EEMD) for signal conditioning and feature extraction, followed by SVM with a radian biases kernel for classification. In the paper [30], SVM, along with, Modular Probabilistic Neural Network (MPNN) and S-transform for feature extraction were explored for the classification of islanding and PQDs in a grid-connected hybrid power system with distributed generation (DG). Eighteen types of power disturbance signals have been classified using an SVM classifier in the study [31]. Disturbance signals, generated via MATLAB simulations, were transformed into images using phase-space reconstruction (PSR) and then into feature vectors. White Gaussian Noise (WGN) was also added to simulate real conditions, with varying SNR levels from 20 to 60. The overall accuracy was reported as 74 %. A novel algorithm using Directed Acyclic Graph Support Vector Machines (DAG-SVMs) and Double-resolution S-transform (DRST) has been proposed for detecting and classifying PQDs [32]. DRST improves feature extraction by enhancing time-frequency localization and reducing computational complexity. The reported accuracy for different types of PQD was above 90 % for each.

4.1.2. Without the incorporation of DSP techniques

The paper [33], presented a method for classifying 5 types of PQDs using variational mode decomposition (VMD) and classification algorithms. Disturbances were generated in MATLAB/SIMULATION and decomposed into eight intrinsic mode functions (IMFs) using VMD. The Random Kitchen Sink (RKS) algorithm, employing an RBF kernel, achieves a classification accuracy of 94.44 %. In contrast, the SVM with an RBF kernel gave an accuracy of 83.33 %.

4.2. Decision tree

A decision tree is a supervised learning method frequently employed in data mining. The classifier takes the form of a tree-like structure consisting of if-else conditions. The top node is known as the root node, and as you navigate through the tree branches and internal nodes, you ultimately reach the leaf nodes. Each node corresponds to a feature or attribute, while each branch emanating from a node represents a decision rule based on that feature.

Gini Impurity quantifies the likelihood of a randomly chosen element being incorrectly labeled if classified according to the distribution of labels in the set. It reaches its minimum value of 0 when all instances in the node fall into the same target category. Let there be classes with relative frequencies p_i , where $i \in \{1, 2, 3, 4, \dots, N\}$. The probability of choosing an item with the label is and the probability of mis-categorizing that item is. The Gini impurity is computed by summing pairwise products of these probabilities for each class label:

$$I_G(p) = \sum_{i=1}^N \left(p_i \sum_{k \neq i} p_k \right) = \sum_{i=1}^N (p_i(1 - p_i)) = \sum_{i=1}^N (p_i - p_i^2) = \sum_{i=1}^N p_i - \sum_{i=1}^N p_i^2 = 1 - \sum_{i=1}^N p_i^2 \quad (11)$$

Entropy is the amount of information needed to accurately describe the sample. For example, if all elements are similar then entropy is 0 else it is a maximum 1. It is defined as

$$H(T) = I_E(p_1, p_2, \dots, p_J) = - \sum_{i=1}^N p_i \log_2 p_i \quad (12)$$

Where are fractions that sum to 1, representing the percentage of each class present in the child node resulting from a splitting the tree.

Algorithm 1, 2, 3.

4.2.1. With the incorporation of DSP techniques

The paper [34] introduces a real-time method for classifying PQD using RMS and discrete Fourier transform (DFT) to extract five key time-frequency features. A rule-based decision tree (RBDT) is utilized to identify nine types of PQDs without the need for a complex classifier. The time-frequency contours for feature extraction have been obtained with Fast ST involving a modified Gaussian window [35]. A hybrid Chemotactic Differential Evolution Algorithm (CDEA), combining BFOA with Differential Evolution (DE) was also presented. This significantly improved power signal pattern classification, achieving over 99 % recognition accuracy even in noisy conditions. The study [36] made significant contributions by introducing fast discrete S-transform (FDST), which further extracts time-localized spectral characteristics accurately with reduced computational cost. It further classifies various PQDs using an automated decision tree (DT) construction algorithm. The research [37] presented a PQD recognition method using a multi-resolution S-transform and decision tree. This approach achieved an accuracy of 99.08 % even in noisy conditions (30db SNR). Stockwell transform and a rule-based decision demonstrated a robust performance of 98 % in recognizing various PQDs like spike, flicker, impulsive transient, etc. [38].

4.2.2. Without the incorporation of DSP techniques

The research [39] introduced a method combining VMD and Decision tree algorithms for detecting and classifying single and complex PQDs in grid-connected DG systems. VMD proves effective in discriminating stationary (harmonics, flicker, and inter-harmonics), non-stationary events (transients), and noise. Utilizing synthetic, real event, and real-time digital simulator (RTDS) generated signals, the method demonstrated above 97 % accuracy for single and mixed PQDs.

Algorithm 1

Random forest for classification and regression.

1. Start with the root node denoted as S which encompasses the entire dataset.
2. Recognize the highest revealing attribute from the dataset using Attribute Selection Measure (ASM), such as information gain and gini index.
3. Split the data S into subsets which contain the possible values having the best attributes.
4. Create a decision tree node that represents the chosen attribute.
5. Apply the same process recursively to each subset generated in step 3 i.e. for each subset, repeat the steps to determine the best attribute, partition the subset, and create corresponding child nodes.
6. Continue the recursion until a stage is reached where further classification of the nodes is not possible. The final node is referred to as a leaf node.

Algorithm 2

Random forest for classification and regression.

1. Repeat for each tree from $b = 1$ to B

- (a) Randomly select a sample of size N (with replacement) from the original dataset.
- (a) For each terminal node, repeat the following step until the node size is less than the minimum number of samples.
 - i. Randomly select m variables from all the p variables
 - i. Pick the most appropriate variable among the m , to separate the data
 - i. Divide the node into two smaller nodes

2. After creating all ' B ' trees, ensemble them $\{T_b\}_1^B$.For the final prediction at a new data point x :

$$\text{Regression : } \hat{f}_{rf}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

Classification: Let $\hat{C}_b(x)$ be the class prediction of the b th random-forest tree. Then

$$\hat{C}_{rf}^B(x) = \text{majority vote} \{ \hat{C}_b(x) \}_1^B$$

Algorithm 3

Random forest for classification and regression.

1. Initialise y for the class variable and x as the feature set.

$$x = (x_1, x_2, x_3, x_4, \dots, x_n)$$

2. Substitute x in the bays theorem and expand using the chain rule.

$$P(y|x_1, \dots, x_n) = \frac{P(x_1|y)P(x_2|y) \dots P(x_n|y)P(y)}{P(x_1)P(x_2) \dots P(x_n)}$$

3. For all entries in the dataset, the denominator $P(X)$ in the above equation remains constant. Thus, it can be omitted from the calculation therefore a proportionality can be introduced.

$$P(y|x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i|y)$$

4. In cases where the classification involves multiple features, the class y with maximum probability is determined.

$$y = \text{argmax}_y P(y) \prod_{i=1}^n P(x_i|y)$$

4.3. Random forest

Bagging or bootstrap aggregation is primarily used to reduce the variance of machine learning models, particularly those prone to overfitting, such as decision trees. For regression, bagging involves training the same regression trees multiple times using random subsets of the training data. The final prediction is obtained by averaging the results from all these trees. For classification, multiple decision trees are trained in the same way, and each tree casts a vote for the predicted class. The class with a majority of the votes is chosen as the final prediction. The technique that combines many weak classifiers to provide solutions to complex problems is known as random forest. It consists of many decision trees. Introduced by Breiman in 2001, this technique uses an advanced version of bagging. Random Forest is widely used because of their simplicity, ease of training, tuning, and effectiveness.

Overall, Random Forest provides a powerful and versatile machine-learning technique that combines multiple decision trees to achieve high accuracy and reduce overfitting. It excels in both classification and regression tasks. Most importantly it can handle diverse datasets and provide insights into feature importance making itself a popular and effective choice in many applications.

In the field of power quality, the random forest technique has been relatively underexplored, with only a few studies leveraging it. Despite its potential, it remains a largely untapped resource in this field of research. The author [40] introduces an algorithm for the detection and classification of PQD using a modified optimal fast discrete ST combined with an RF classifier and achieves 99.61 % accuracy. In the proposed algorithm, modified ST effectively enhances time-frequency localization with improved energy concentration while the RF classifier helps in reducing variance and bias. The paper [41] presents a novel approach for optimal feature selection and classification using a wavelet-based random forest classifier. The method significantly increases the accuracy to 98.71 for 7-type disturbances and 98.76 % for 10-type disturbances. The authors [42] introduced an entropy-importance (EnI) based random forest model for selecting features and classifying power quality disturbances. Using 35 features extracted from S-transform data with random noise, it quantifies feature importance using EnI, excluding irrelevant features and selecting the optimal feature sub-set with sequential forward search (SFS). The proposed model achieved high classification accuracy of 99.7 % at 30 dB and 96.8 % at 20 dB.

4.4. Fuzzy system

A fuzzy system consists of four main components namely, rule base, fuzzification, interference engine, and defuzzification. The rule base contains the set of rules, if-then conditions, and membership functions used by the system. Fuzzification or a fuzzifier is used to convert precise input data into fuzzy values by using predefined membership functions. The third component, the interference engine, evaluated how well the current fuzzy inputs match each rule. Once the relevant rules are triggered, the inference engine combines them to generate the appropriate control actions or outputs. The last component, defuzzification or de-fuzzifier transforms fuzzy results back into precise values for actionable outputs.

4.4.1. With the incorporation of DSP techniques

The paper [43] introduces a wavelet-based extended fuzzy reasoning approach for recognizing and identifying PQDs. The rule basis for fuzzy reasoning is created using wavelet energy distributed features. The approach maps disturbance features to real values for accurate identification and shows zero error rates in tests. A modified Fuzzy C-means algorithm enhanced by bacterial foraging optimization technique (BFOA) has been used to improve classification accuracy [44]. The study classifies non-stationary power signals using EMD combined with a short-time modified Hilbert transform (STMHT). The proposed method achieved 99.63 % of classification accuracy. The research [45] used an S-transform-based fuzzy expert system (FES) for mining power quality time series data. To achieve high classification accuracy, a fuzzy logic-based adaptive particle swarm optimization (PSO) technique is utilized to optimize the membership function.

4.4.2. Without the incorporation of DSP techniques

Modified fuzzy min-max (MFMN) clustering neural network is demonstrated as an effective technique for power quality monitoring and data clustering [46]. It can handle dynamic and noisy data and offers robust performance.

4.5. Bayesian classifier

A Bayesian classifier uses probabilities to determine which class or category an observation or data belongs to. It applies the Bayes theorem, which relates the likelihood of a class given the observed features. To the prior probabilities of the class and the likelihood of the features given the class.

Given an example with inputs $X_1 = v_1, \dots, X_k = v_k$, Bayes' rule is utilized to compute the posterior probability distribution of the example's classification, Y .

$$\begin{aligned} P(Y|X_1 = v_1, \dots, X_k = v_k) &= \left(P(Y|X_1 = v_1, \dots, X_k = v_k) \times \frac{P(Y)}{P(Y|X_1 = v_1, \dots, X_k = v_k)} \right) \\ &= \left(\left(\frac{P(X_1 = v_1, \dots, X_k = v_k)}{Y} \right) \times P(Y) \right) \left(\sum_Y ((P(X_1 = v_1, \dots, X_k = v_k) / Y) \times P(Y)) \right) \end{aligned} \quad (13)$$

The study [47] used a three-level multiply connected Bayesian network (PLBN) for detecting and classifying complex power PQDs. The model consisted of three layers features evidence layer, circumstance evidence layer, and disturbances state layer. It outperforms other methods by demonstrating using simulation and real-life tests. The paper [48] examines the classification of PQ events using 8 different classification methods, including Bayesian and SVM classifiers. The research [49] introduced a fast Time-Time Transform for analyzing and recognizing patterns in non-stationary signals. The features were then classified using the naïve Bayes classifier.

4.6. Rule-based classifier

A rule-based classifier is a classification algorithm that employs a set of "if-then" rules to make predictions based on the input data. The rule base contains multiple rules that cover various scenarios. Fig. 5 illustrates a general rule-based classifier tree and its corresponding rule set. The figure provides an example featuring 4 nodes and 5 associated rules.

The paper [50] presents a method that uses a rule-based classification approach for PQD classification and S-Transform for feature extraction. It employed linear and parabolic rules with decision boundaries set through a heuristic approach. A real-time rule-based system was also developed for classifying PQDs using S-Transform features [51]. It also employed an advanced digital system processor for fast signal processing.

4.7. Neuro-fuzzy system

Neuro Fuzzy Systems (NFS) are hybrid systems that combine fuzzy logic with neural network learning which determine fuzzy sets and rules by processing data samples through algorithms inspired by neural networks. They can be visualized as a 3-layer feed-forward

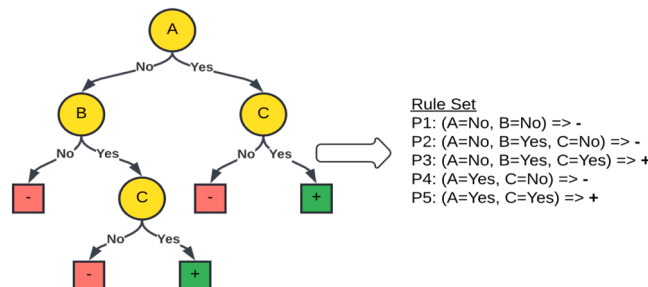


Fig. 5. An example of a Decision Tree for a Rule-Based Classifier.

neural network wherein, the input layer represents input variables, the fuzzy rules are described using a hidden layer, and the output variables are represented by the output layer. This system can be interpreted as a collection of fuzzy rules and can be created from training data or initialized with prior learning. Fig. 6 depicts the basic algorithm behind the neuro-fuzzy system.

The paper [52] presents a novel approach for detecting and classifying PQ events using a four-step automatic algorithm the method involves representing the voltage signal in a 3-D space applying principal component analysis (PCA) to extract features from eigenvalues of disturbance waveforms and using a neuro-fuzzy classifier for automated classification. The study in [53] addresses PQ issues in DG systems by using the Hilbert-Huang transform for extraction of features from power quality signals, these features are then classified using an adaptive NFS. Another study [54] addresses the challenge of detecting PQ events using a wavelet transform coefficient, it further employs a neuro-fuzzy classification system for constructing fuzzy rules and recognizing signals effectively. Table 3 presents the training and inferences complexity of various ML algorithms which is important to consider for computationally efficient algorithms. In the table n denotes samples, m denotes dimensions, c denotes classes and d_{tree} denotes the depth of the tree structure.

5. Automatic classification of PQDS using deep learning techniques

In the late 1980s, neural networks gained prominence in the fields of ML and AI because of the growth of efficient learning algorithms and innovative network architectures [55]. A “new-generation neural network” occurred when Hinton et al. introduced the concept of deep learning in 2006, revitalizing the interest in neural networks [56]. Presently, deep learning has become increasingly prominent due to advancements in computational power, the availability of data in abundance, and the development of innovative algorithms and network architectures [57]. AI encompasses the broader development of machines and algorithms that replicate human intelligence by optimizing actions to achieve specific objectives. Deep Learning (DL) and Machine Learning (ML) are the two subsets of AI. While ML focuses on enabling systems to learn from data without explicit programming, DL utilizes multilayer neural networks to learn from huge amounts of data and address complex tasks. DL, being at the forefront of AI’s evolution underscores its essential role in advancing intelligent systems and providing solutions to intricate problems across numerous domains including healthcare, autonomous systems, finance, cybersecurity, environmental science, manufacturing and industry, and many more. The main algorithms in deep learning include Convolutional Neural Network (CNN), Restricted Boltzmann Machine (RBM), Deep Belief Network (DBN), Recurrent Neural Networks (RNN), Generative Adversarial Network (GAN), Transformers, Autoencoders, etc.

The process of utilizing deep learning for an application includes several key steps. The approach starts with data understanding and preprocessing. It begins with data understanding and preprocessing, which includes collecting real-world data, annotating it, and preparing it for analysis. The next step is building and training the deep learning model, which involves leveraging advanced algorithms and computation resources. It also involved parallelized matrix and tensor operations. The final step is model evaluation and interpretation.

Besides its major role in power quality monitoring, deep learning techniques have a wide range of applications in diverse fields such as medical [58–60], agriculture [61,62], human emotion and speech recognition [63,64], climate and environmental monitoring [65, 66], banking [67], fraud detection [68,69], road safety [70] and more. To understand the concept of deep learning, it is required to grab some fundamental concepts that play an important role in the model’s performance. A neural network is the basic building block of deep learning. It consists of interconnected nodes or neurons organized into a single input layer, multiple hidden layers, and a single output layer. A neuron is a fundamental unit of a neural network. It receives multiple inputs; each assigned a weight that regulates how much influence each input has during the forward pass. In this process, the neuron computes the weighted sum of these inputs and then applies a transformation function to this sum to produce the output. The weight of a neuron refers to the value telling the neuron how much to pay attention to each piece. The computation of a neuron can be understood as follows [4]. As per [4], consider a neuron z_i at a layer l , receiving an input vector $a_j(l-1)$, the computation of the neuron is defined by,

$$z_i^l = \sum_{j=0}^d w_{i,j}^{l-1} \cdot a_j^{l-1} \quad (14)$$

$$a_i^l = g(z_i^l) \quad (15)$$

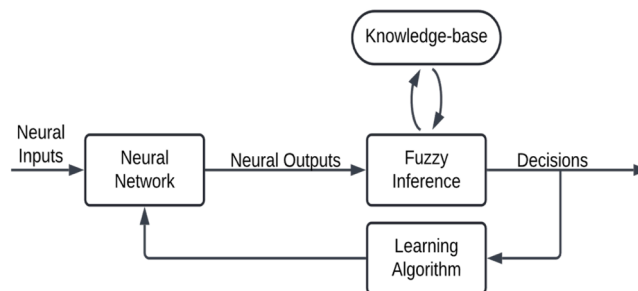


Fig. 6. Fundamental Algorithm of a Neuro-Fuzzy System.

Table 3
Training and Inference Complexity of Various Machine Learning Algorithms.

ML Algorithms	Training Complexity	Inference Complexity
Support Vector Machine	$O(n^2m + n^3)$	$O(mnsv)$
Decision Tree	$O(n \log(n).m)$	$O(dtree)$
Random Forest	$O(n^2m)$ *worst case	$O(ntrees.dtrees)$
Bayesian Classifier	$O(ntrees.n \log(n).m)$	$O(mc)$

Where g is the transformation function or the activation function and w_{ij} is the set of weights between the inputs to the neuron. A convolutional neural network (CNN) is a specialized form of neural network mainly used for image recognition and classification tasks. CNNs are designed to automatically detect patterns such as edges, textures, and shapes within images by using convolutional layers, followed by pooling layers. This architecture makes CNNs highly effective for processing visual and spatial data, making them widely used in computer vision tasks.

An Artificial Neural Network (ANN) is a more general-purpose neural network architecture used for pattern recognition and data classification. ANNs consist of layers of interconnected neurons, where each neuron processes input data using weights and activation functions. The data is passed through multiple layers, including input, hidden, and output layers, to make predictions or classifications. While ANNs can be applied to a wide variety of tasks, they are less specialized than CNNs for images or RNNs for sequential data.

A Recurrent Neural Network (RNN) is designed to deal with sequential data, such as speech, time series, and text. RNNs have connections that enable them to retain information from previous inputs, allowing the network to maintain a memory of earlier data points in the sequence. This makes RNNs well-suited for tasks like language modeling and sequence prediction. However, traditional RNNs can struggle with long-term dependencies, which more advanced variants, like Long Short-Term Memory (LSTM) networks, address.

5.1. Hyperparameters and evaluation metrics of a deep learning model

5.1.1. Activation function

Activation Functions are crucial non-linear components in neural networks that enable the composition of several building blocks for the learning of complex nonlinear functions. They convert the weighted sum of inputs into a more classifiable form by introducing non-linearity, allowing the network to capture complex relationships between inputs and outputs. Some common activation functions include sigmoid, softmax, hyperbolic tangent (tanh), rectified linear unit (ReLU), leaky ReLU, Swish, etc. The sigmoid activation function maps the real numbers to the range (0, 1). Softmax activation function converts a vector of raw prediction scores into a probability distribution, and it is typically used in multiclass classification tasks. Rectified Linear Unit (ReLU) outputs the input directly if it is positive and returns zero otherwise. It is primarily used in the hidden layers of neural networks. The mathematical representations of the sigmoid, softmax, and ReLU activation functions are shown in Eqs. (16), 17, and 18 respectively.

$$S(x) = \frac{1}{1 + e^{-x}} \quad (16)$$

where,

$S(x)$ = sigmoid function

e = Euler's number

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (17)$$

where,

$\sigma(\vec{z})$ = softmax function

e^{z_i} = standard exponential function for an input vector

$\vec{z}$ = input vector

K = number of classes in the multi-class classifier

e^{z_j} = standard exponential function for output vector

$$ReLU(x) = \max(x, 0) \quad (18)$$

where,

$ReLU(x)$ = Rectified linear unit

x = input to the function

5.1.2. Loss function

A loss function, often referred to as a cost function quantifies the extent to which a deep learning model's predictions align with the actual target values. An optimization algorithm called gradient descent is widely used to optimize the weights of deep learning models. It optimizes the weights by iteratively updating them to minimize the prediction error across all instances in the training set. The prediction error is what is a loss [71]. The loss is quantified by a loss function. The weight update is defined as

$$w_{i,j}^{l-1} = w_{i,j}^{l-1} - \delta \nabla_{w_{i,j}} \quad (19)$$

where $\nabla_{w_{i,j}}$ denotes the gradient of the loss function J with respect to the weight $\frac{\partial J}{\partial w_{i,j}}$ and δ is a hyperparameter called the learning rate.

The binary cross entropy is typically used for binary classification tasks and categorical cross entropy is applied to multiclass classification tasks. For regression tasks, mean absolute error (MAE) and mean squared error (MSE) are the two commonly used loss functions. The expressions for the two are mentioned below.

$$MSE = \frac{1}{n} \sum_{i=0}^n (y_i - \hat{y}_i)^2 \quad (20)$$

$$MAE = \frac{1}{n} \sum_{i=0}^n |y_i - \hat{y}_i| \quad (21)$$

where,

n is the number of observations

$\hat{y}_i$ is the predicted value

y_i is the true value

5.1.3. Learning rate

The learning rate is a critical hyperparameter determining how much the model's weights are updated during each step of learning. Adjusting the learning rate appropriately significantly improves both training efficiency and generalization. A larger learning rate causes the model to overshoot the optimal solution, resulting in instability and poor convergence. While an excessively smaller learning rate unnecessarily takes longer time in training [72]. The crucial factors that play a crucial role in deciding the learning rate include model complexity, size of the dataset, and batch size.

5.1.4. Epochs and batch size

An epoch is defined as one complete pass through the entire training dataset. The training process of the model involves running multiple epochs to allow the model to learn better representations and improve its performance. Factors influencing the number of epochs include model convergence to a stable state, model overfitting, and training time. Another hyperparameter, batch size, refers to the number of training samples used in one iteration of updating the model's weights. Epochs and batch sizes are decided keeping in mind the dataset size, model complexity, and hardware limitations.

5.1.5. Evaluation metrics

Deep learning models are assessed using various evaluation metrics, including Accuracy, Precision, Recall, Specificity, Negative Predictive Value, and F1-Score. These metrics provide insights into the model's performance and accuracy. The formulas for these standard evaluation metrics are summarized in Table 4, with the notations defined as follows: TP represents True Positive, TN denotes True Negative, FP stands for False Positive, and FN signifies False Negative.

Table 4
Formulae for Standard Evaluation Metrics.

Evaluation Metric	Formulae
Accuracy	$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$
Precision	$Precision = \frac{TP}{TP + FP}$
Recall or Sensitivity	$Recall = \frac{TP}{TP + FN}$
Specificity	$Specificity = \frac{TN}{TN + FP}$
Negative Predictive Value	$Negative\ Predictive\ Value = \frac{TN}{TN + FN}$
F1-Score	$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall}$

5.2. With the incorporation of DSP technique

Many researchers have leveraged the power of deep learning to address the challenges of classifying and detecting power quality disturbances (PQDs). Their ability to learn intricate patterns and handle large datasets has proven effective in this domain. The paper [73] proposed the method for classifying PQDs using the Hilbert transform and radial basis function (RBF) neural networks. Hilbert transform has been used for extracting features from disturbance signals and RBF neural network has been employed for classification. The combination provided the highest accuracy of 97 %. The study [74] used a probabilistic neural network with Hilbert Transform to identify voltage sag. The approach decomposed non-stationary signals into Intrinsic Mode Functions (IMFs) using Empirical Mode Decomposition (EMD). EMD with PNN proved to be more effective with 98.63 % efficiency. The research [75] highlighted automatic power quality event recognition using an Extreme Learning Machine (ST-ELM) and ST. The study used 10 different classification procedures to evaluate the performance of the ELM classifier, demonstrating the high performance of the ST-ELM-based system. Multi-layer perceptron ELM with empirical wavelet transform has been utilized for PQD classification [76]. It employed hierarchical ELM (H-ELM) enhanced by particle swarm optimization (PSO) for feature selection. The study achieved 93 % of overall accuracy.

The paper [77] presented a novel approach for PQF detection and classification using a radial basis function neural network (RBFNN) combined with wavelet transform. The approach can classify 20 types of disturbances and achieve 97.85 % classification accuracy. Extreme Learning Machine (ELM), a single hidden layer neural network known for its fast learning and good generalization capabilities, has been largely explored in the study [78]. The study introduced a PSO-based wrapper model to determine the optimal number of hidden nodes and their effective selection, thereby improving the ELM's performance in PQD classification. It achieved overall classification accuracy of 97.60 %. The study [79] used an Artificial Neural Network (ANN) with empirical mode decomposition (EMD) addressing the challenge of low signal-to-noise ratio (SNR). The methodology demonstrated high effectiveness, achieving 100 % accuracy. The research [80] utilized the combination of Discrete Wavelet Transform (DWT), Probabilistic Neural Network (PNN), and Artificial Bee Colony (ABC) optimization for the automatic classification of single and multi-stage PQDs. The paper [81] presented a methodology using a generalized S-Transform with an optimized width factor and a probabilistic neural network (PNN). It classified eight disturbance signals and one normal signal, resulting in a classification accuracy of 99.26 % without noise, 99.13 % with 40 dB noise condition, and 98.63 % with high-level noise condition (30 dB).

The study [82] presented a novel method using singular spectrum analysis (SSA) for recognizing weak transient signals, curvelet transforms (CT) for efficient signal decomposition, and deep convolutional neural networks (DCNNs) for classification. The SSA-FDCT-DCNN (SFD) outperformed even multiclass SVM. The research [83] investigated fully complex-valued radial basis function networks (FC-RBF) and complex ELM. It utilized S-Transform for feature extraction and the extracted features form input to the FC-RBF and CELM network. The testing accuracies for various classes range from 90–100 for both FC-RBF and CELM. An entirely new methodology has been introduced in [84], utilizing a combination of compressed sensing (CS) and sparse autoencoder (SAE) techniques. The first one extracts features from PQD waveforms while the other reduces redundancy and enhances classification performance. Nine single and twenty-two combined PQD classes have been classified with backpropagation fine-tuning of the network for optimal accuracy. 99–100 % accuracy has been achieved using the aforementioned method. The paper [85] proposed a method leveraging compressive sensing (CS), singular spectrum analysis (SSA), wavelet transform (WT), and deep neural networks (DNNs). The SSA-CS-DNN algorithm was tested on 15 types of single and combined PQDs and demonstrated outstanding performance of 99.855 % and 99.63 % accuracy without and with noise respectively. Another research [86] utilized the combination of compressed sensing (CS) and deep CNNs. CS reduces the demand for data collection and enhances the transmission efficiency by compressing PQD signals.

A hierarchical ELM (H-ELM) has been utilized in the study [87]. Alongside this, it employed discrete wavelet transform (DWT) for feature extraction, PSO for optimal feature selection, and H-ELM classifier for robust accuracy with minimal parameter adjustment. A hybrid approach has been introduced in [88] combining phase space reconstruction (PSR) with CNN for PQD classification. The methodology involved transformation from 1D voltage disturbance signals to 2D images using PSR. With automatic feature extraction, the proposed method achieved 99.8 % accuracy. Another automatic feature extraction was proposed in [89] using Wigner-Ville distribution (WVD) and CNN. Class activation maps (CAM) were also used to visualize and analyze classification errors and the approach achieved 99.67 % accuracy. The paper [90] introduced a combined approach using 2D-RT signal processing and Multi-Objective Grey Wolf Optimizer MOGWO with the k-nearest Neighbor (KNN) algorithm. This 2D-RT-MOGWO-KNN method achieved 99.81 % accuracy in noisy conditions and 97.62 % on experimental data. The study [91] analyzed synthetic PQ signals in the time-frequency domain using hyperbolic window Stockwell transform (HWST) and performed classification using the XGBoost algorithm. Deep features were extracted from the transformed signals using a 3-layer stacked autoencoder (SAE). The overall reported accuracy was 99.86 %. In the paper [92], segmented and modified S-Transform (SMST) was used for providing high-resolution 2D time-frequency maps of PQ signals, deep CNN to automatically extract features, and multiclass SVM was used for classification. The proposed method achieved classification accuracies of 99.7 %, 98.86 %, 99.52 %, and 99.6 % with 0 dB, 20 dB, 30 dB, and 40 dB noise respectively. A deep stacking network (DSN) has been used for classification, with enhancement by the Batch-mode gradient fine-tuning method, in the study [93]. Along with this, the research also utilized discrete orthogonal S-transform (DOST) for feature extraction and compressive sensing (CS) for feature reduction. The paper [94] brought novelty by combining Stockwell Transform (ST) with an improved Grey Wolf Optimization-based Kernel Extreme Learning Machine (IGWO-KELM). The IGWO optimized the parameters of the KELM model, and the approach achieved an accuracy of 99.76 % under noiseless conditions and 97–98 % accuracy at different SNR levels.

The study [95] proposed a hybrid feature selection method based on correlation evaluation and Genetic Algorithm-Extreme learning machine (GA-ELM) for optimal feature reduction. The approach achieved over 99 % accuracy, surpassing single-classifier methods in reliability. The paper [96] proposes a discrete wavelet transform (DWT) 1D-CNN approach for compressing and

classifying PQDs. By leveraging DWT and multi-resolution analysis (MRA), it enhances feature extraction, while CNN automates classification. The paper [97] presents a novel PQD classification approach by transforming voltage signals into colorized wavelet coefficient images and applying CNNs, eliminating traditional feature selection steps. The model is validated on real-life disturbance signals from Turkey, achieving 99.8 % accuracy. This study [98] proposes a deep learning-based PQD detection system using Stockwell Transform (ST) contour images as input to a CNN optimized with Bayesian Optimization (BOA). The approach enhances feature extraction and classification performance for PQDs in hybrid power systems and achieved over 99 % accuracy in all cases. This study [99] introduces a PQD classification approach using Histogram of Oriented Gradients (HOG) for feature extraction and Extreme Learning Machine (ELM) for classification. This method enhances computational efficiency while maintaining high classification accuracy. The study [100] proposes a hybrid model integrating wavelet transform, attention mechanism, and LSTM for PQD classification. The approach enhances spatial and temporal feature extraction for improved classification accuracy achieving 89.77 % With highest noise.

5.3. Without the incorporation of DSP techniques

The research [101] presented a novel methodology using a dual neural network approach without relying on DSP techniques. It consists of two neural networks: an adaptive linear network (ADALINE) for the estimation of harmonic and inter-harmonics and a feed-forward neural network (FFNN) for pattern recognition. The ADALINE also computes root mean square value of voltage and total harmonic distortion (THD) indices. It detected and classified 96 nonmutual exclusive combinations of PQ disturbances even if they occur simultaneously or in noisy environments. ADALINE was outperformed by the variational mode decomposition (VMD) and a deep stochastic configuration network (DSCN) [102]. The method used VMD for handling the time-frequency resolution of both stationary and non-stationary events and extracting statistical features. It used DSCN to make the model learn faster. The duo of VMD with DSCN achieved 99.4 % accuracy. The CNN network proposed in [103] demonstrated higher accuracy than ResNet50 and LSTM networks. The model effectively identified single and compound PQ disturbances using a single closed-loop feedback training process. It excelled with 99.96 % classification accuracy for classification among 16 PQDs. The study [104] detected and classified PQDs in a distribution network with distributed generation. VMD has been used to filter noise and extract mode functions, which were then analyzed using Detrended Fluctuation Analysis (DFA). The proposed algorithm was able to classify nine PQDs and achieve 99.38 % accuracy under 30 dB noise, 96.75 % accuracy under 20 dB noise, and 94.82 % accuracy under 10 dB noise. To mitigate the issues caused by PQDs, it is essential to determine their type and the starting and end times.

The study [105] introduced a deep learning approach using a stacked Long Short-Term Memory (LSTM) model combined with a sliding time window. The method involved transforming the collected PQD wave into a grayscale image which is then fed into the LSTM model. The reported mean accuracies are 99.64 %, 99.88 %, and 99.91 % with 20 dB, 30 dB, and 50 dB SNR values. K-means clustering combined with pattern recognition has been used effectively to cluster and identify power quality events and interruptions [106]. The approach [107] used a linear mathematical decision-based rule for predicting and classifying power quality issues, especially in power grids with increased renewable energy integration. The research [108] introduced a novel hybrid CNN method using both 1D and 2D CNN architectures. The 1D CNN processes power signals in their original form and then converts them into images for analysis using 2D CNN. The combined feature vectors of the two architectures along with a fully connected layer, have been used to classify PQDs. The proposed method achieved 99.97 % accuracy, 99.99 % specificity, and 99.95 % sensitivity. The paper [109] introduced an advanced deep convolutional network, especially for addressing the issues of low convergence speed, accuracy, and generalization in traditional methods. The network combined a 5-layer 1D modified Inception-residual network (1D-MIR) with a 3-layer fully connected tier. The network completed the training in approximately 6 min and attained a high classification accuracy of 99.1 %.

The paper [110] proposed the Single Shot Power Quality Disturbance Detection (SSPQDD) architecture, especially for multiple disturbance scenarios. The architecture achieved 96.55 % accuracy. The study [111] used CNN's ability combined with an attention model. The method involved signals from the dataset with 9 classes, each characterized by 622 data points and 5600 parameters. This approach achieved a high accuracy of 99.92 %. The paper [112] integrated CNNs with LSTM networks to enhance the detection process. The 1D CNN extracts the spatial features and then the LSTM network extracts the temporal features. The proposed duo achieved 100 % accuracy in the case of pure signal and signal with 40Db noise, 99.69 % and 99.31 % accuracy with 30 dB and 20 dB noise respectively. The study [113] demonstrated the use of the LSTM neural network for voltage sag subtype classification and achieved up to 92.68 % accuracy on real-world datasets. The research [114] utilized Deep Auto-encoder (DAE) networks, which use the Gabor filter to extract general features from PQD images and a sparse-based DAE network to identify and learn the crucial features. The method excels in achieving 99.4 % accuracy and covering 15 PQ events in its classification.

The study [115] has proposed a Transformer deep learning framework capable of directly processing voltage signals and accurately classifying PQEs under challenging conditions, such as measurement noise, DC offsets, and variations in amplitude and frequency, achieving accuracy between 99.81 % and 91.43 % across varying conditions. The study [116] highlights the effectiveness of Vision Transformers (ViT) in outperforming CNNs for image classification tasks when trained on sufficient data. The 2D image was generated from 1D signals and passed through the vision transformer model for further classification. Table 5 summarises the prior research on machine learning methods used for the PQD classification and on the other hand Table 6 summarises the Deep learning methods for PQDs classification. The SmartArt chart in Fig. 7(a) illustrates the distribution of reviewed papers categorized by the amount of literature employing various ML and DL techniques while Figure 7(b) presents a comprehensive analysis of the papers reviewed by the authors, highlighting the percentage distribution across various publishing houses.

Beyond the progress in the classification of PQDs, extensive research has been carried out in the field of power quality enhancement

Table 5
Summary of Machine Learning-based PQD Classification Research.

Ref.	AI Technique	DSP Technique	Accuracy	Advantages	Disadvantages	Dataset Type
[24]	Support Vector Machine	Wavelet Transform	Pure signal – 99.43 % Signal with 30 dB noise – 98.63 % Signal with 20 dB noise – 97.75 %	The proposed approach achieves high classification, accuracy, and robustness, particularly when applied to both noisy and noiseless signals. It optimizes memory usage and computational efficiency through feature selection and wavelet transformation techniques.	The computational cost increases during the training phase due to the feature selection and cross-validation processes. Performance can degrade when lower decomposition levels or simpler wavelets (e.g., db2) are used to reduce computation time further.	Synthetic
[25]	Least Squares Support Vector Machine	Wavelet Transform	98.88 %	The proposed k-means-based Apriori algorithm improves the efficiency of feature selection, resulting in high classification accuracy. The system is reliable, and applicable, and processes large datasets effectively, using multiresolution analysis and LS-SVM classifier.	The training process, involving 10-fold cross-validation and feature extraction, may increase computational overhead. The reliance on k-means clustering might reduce performance when dealing with non-linear data separations.	Synthetic & Real
[26]	Support Vector Machine for Classification of Voltage Disturbances	-	-	The binary decision tree structure, with individually trained SVMs as nodes, allows for straightforward extension when new disturbance types are added.	The system relies heavily on real data for optimal performance, which may not always be readily available compared to synthetic data.	Synthetic & Real
[29]	Support Vector Machine	Hilbert-Huang Transform	-	The combination of EEMD and HHT effectively extracts distinctive features like instantaneous frequency (IF) and instantaneous amplitude (IA), improving classification performance.	The hybrid algorithm involves multiple steps (EEMD, HHT, feature extraction, and statistical calculations), making it computationally intensive and potentially difficult to implement in real-time systems.	Synthetic
[30]	Modular Probabilistic Neural Network and Support Vector Machines	S-transform	MPNN – 96.17 % SVM – 97.67 %	The used classifiers allow for the effective detection and classification of both islanding and PQDs, providing high accuracy and reliability.	Implementing two different classifiers (MPNN and SVM) increases the system's complexity, requiring additional computational resources for training and testing both models.	Synthetic & Real
[31]	Support Vector Machines	–	Pure signal – 96 % Signal with 60 dB noise – 88 % Signal with 40 dB noise – 88 % Signal with 20 dB noise – 74 %	The combination of phase space reconstruction and support vector machines (SVM) enables the effective identification and classification of various PQDs. It also proves to be robust under noisy conditions.	Phase space reconstruction for feature extraction may involve complex computations, which can make the method more resource-intensive and challenging to implement in real-time systems.	Synthetic
[32]	Directed Acyclic Graph Support Vector Machines	S-transform	Pure signal – 97–100 % Signal with 20 dB noise – 92–100 %	The combination of Directed Acyclic Graph Support Vector Machines (DAG-SVMs) and Double-Resolution S-Transform (DRST) enhances classification accuracy and robustness in identifying single-stage and complex PQDs.	The selections of parameters (λ_1 and λ_2) used in DRST, while effective in the presented experiments, may require further optimization for better generalization.	Synthetic
[33]	Support Vector Machine and Random Kitchen Sink	-	RKS – 94.44 % SVM – 88.89 %	Variational Mode Decomposition (VMD) effectively decomposes power quality (PQ) disturbances into intrinsic mode functions (IMFs), providing a robust feature extraction method.	The study only focuses on five disturbance types (swell, sag, harmonics, sag with harmonics, swell with harmonics), limiting the generalizability of the approach to a broader range of PQ disturbances like flicker, notch, or glitches.	Synthetic
[35]	Fuzzy decision tree	Fast S-Transform	-	The hybrid Chemotactic Differential Evolution Algorithm (CDEA) improves classification accuracy by overcoming the limitations of premature	The hybridization of BFOA and Differential Evolution (DE) adds complexity to the algorithm design, making it computationally more intensive and harder to implement	Synthetic

(continued on next page)

Table 5 (continued)

Ref.	AI Technique	DSP Technique	Accuracy	Advantages	Disadvantages	Dataset Type
				convergence and slow optimization found in the Bacterial Foraging Optimization Algorithm (BFOA), leading to more precise power quality (PQ) disturbance classification.	in real-time systems without adequate computational resources.	
[36]	Decision Tree	Fast S-Transform	-	The Fast Discrete S-Transform (FDST) variants significantly reduce the computational cost while maintaining high accuracy in extracting time-localized spectral characteristics from nonstationary signals.	Although frequency partitioning schemes reduce computational cost, they add an extra layer of complexity to the signal processing workflow, which may pose challenges for implementation in systems with limited resources.	Synthetic & Real
[37]	Decision Tree	Multiresolution S-Transform	-	The use of Multiresolution S-transform (MST) with controllable adjustment factors enhances the time-frequency resolution, leading to better feature extraction and more accurate recognition of PQ disturbances.	While the method performs well for the tested disturbances, its applicability to a broader range of PQ issues, or those with highly dynamic characteristics, may require further validation.	Synthetic
[40]	Random Forests	Modified S-Transform	-	The use of a Random Forest (RF) classifier, leveraging bagging and decision stumps, enhances classification performance by reducing variance and bias.	The optimization of window parameters in the modified ST introduces additional complexity to the process.	Synthetic
[42]	Entropy-importance-based Random Forests	-	Signal with 30 dB noise or above – 99.7 % Signal under 20 dB noise – 96.8 %	EnI leads to better feature subset selection, which enhances the overall classification accuracy, making the model more effective than traditional methods like Sequential Forward Search (SFS) and Sequential Backward Search (SBS).	Although the method performs well under noisy conditions, it relies on the assumption that noise levels are manageable.	Synthetic
[44]	Fuzzy C-means and Bacterial Foraging Algorithm	Hilbert Transform	99.63 %	The proposed combination of Empirical Mode Decomposition (EMD) and Short-Time Modified Hilbert Transform (STMHT) provides an improved time-frequency distribution for non-stationary power signal disturbances.	Neuro-fuzzy classification systems are often complex to practical problems.	Synthetic
[50]	Rule-based classifier	S-transform	-	The rule-based approach uses a reduced and simple set of features, allowing for easier implementation and faster processing in comparison to more complex methods like neural networks.	While effective for predefined disturbance types, the rule-based system may struggle with new or unaccounted disturbance patterns that it hasn't been explicitly trained for.	Synthetic
[52]	Neuro-fuzzy	Wavelet Transform	-	The integration of a noise-suppression algorithm significantly enhances the recognition rate of power quality (PQ) events, even when the signals are corrupted by noise, improving overall system reliability.	While effective, neuro-fuzzy classification systems can be complex to design and require careful tuning of fuzzy rules, which might not be straightforward for all users.	Synthetic & Real

[117–120]. Despite these advancements, significant research gaps remain, particularly concerning utilizing advanced approaches such as Transformers, multi-feature fusion approaches [121], and dimensionality reduction methods [122], which have shown exceptional performance in other domains.

6. Case study

This case study focuses on evaluating the performance of various ML classifiers in classifying PQDs, complementing the review of studies on diverse AI techniques. This not only adds practical insights but also provides a hands-on investigation, enhancing the overall understanding. Initially, five simple and complex PQDs were generated using mathematical models in MATLAB. The equation and parameters of these PQ disturbances are mentioned in Table 7 as per IEEE-1159 standard [15]. The preprocessing of PQDs involves a triggering process that is utilized for the segmentation purpose which means identifying the beginning and ending of the PQD. For

Table 6
Summary of deep learning-based PQD classification research.

Ref.	AI Technique	DSP Technique	Overall Accuracy	Advantages	Disadvantages	Dataset Type
[73]	Artificial Neural Network (ANN)	Hilbert Transform (HT)	94 %	The RBF network, which is part of the proposed approach, requires less training time compared to other ANN architectures, enhancing the system's overall efficiency.	While the RBF network is highlighted for its speed and accuracy, the paper does not explore the performance of alternative neural networks or hybrid approaches that may offer additional improvements in certain scenarios.	Synthetic
[74]	Probabilistic Neural Network (PNN)	Hilbert Transform	98.63 %	The use of EMD to decompose power quality disturbance signals into Intrinsic Mode Functions (IMFs) is a significant strength.	The paper focuses on only three causes of voltage sags. While these are common in power quality disturbances, the scope could be broadened to include other potential causes, making the model more comprehensive and applicable in a wider range of scenarios.	Synthetic
[75]	Extreme Learning Machine (ELM)	S-Transform	-	The proposed ST-ELM approach demonstrates very high performance in classifying both real and synthetic PQ event data. The method's success in dealing with both types of datasets shows its robustness.	The paper does not provide a deep analysis of how S-Transform compares to other feature extraction methods (such as wavelet transform) under similar conditions. A comparative study could strengthen the claim that ST is the best choice for this application.	Synthetic
[76]	Hierarchical Extreme learning machine (H-ELM)	Discrete Wavelet Transform	93 %	By using PSO for feature selection, the model ensures that only the most relevant features are used for classification, leading to better accuracy and reduced computational overhead.	Although the method is efficient, the combination of DWT, PSO-based feature selection, and H-ELM could introduce additional complexity in terms of implementation. This might require more effort during system setup and parameter configuration.	Synthetic
[77]	Radial basis function neural network	Wavelet Transform	97.85 %	The paper introduces an enhanced version of RBFNN, where the weights are updated using Particle Swarm Optimization (PSO). The cognitive and social behavior of particles, along with a fitness function, is used to optimize the weight update process, improving the network's performance.	The paper also introduces an enhanced version of RBFNN, where the weights are updated using Particle Swarm Optimization (PSO). The cognitive and social behavior of particles, along with a fitness function, is used to optimize the weight update process, improving the network's performance.	Synthetic
[78]	Extreme Learning Machine (ELM)	Wavelet Transform	97.60 %	The Extreme Learning Machine (ELM) combined with Particle Swarm Optimization (PSO) offers a highly efficient approach for the multicategory classification of PQDs.	Incorporating PSO for feature selection introduces computational overhead, which can increase the time required for model training and optimization.	Synthetic
[79]	Artificial Neural Network (ANN)	Empirical Mode decomposition	100 %	By integrating Empirical Mode Decomposition (EMD) with parameter estimation techniques like Shannon entropy, skewness, and kurtosis, and leveraging Artificial Neural Networks (ANNs), it achieves exceptional classification performance.	By integrating Empirical Mode Decomposition (EMD) with parameter estimation techniques like Shannon entropy, skewness, and kurtosis, and leveraging Artificial Neural Networks (ANNs), it achieves exceptional classification performance.	Synthetic
[80]	Probabilistic Neural Network-based Artificial Bee Colony (PNN-ABC)	Discrete Wavelet Transform	99.875 %	The ABC algorithm ensures the selection of optimal features and the best spread constant, which helps in eliminating redundancy and improving the classifier's performance.	The approach's reliance on optimal feature selection means that its success is highly dependent on the quality and representativeness of the feature set used, which could affect its	Synthetic

(continued on next page)

Table 6 (continued)

Ref.	AI Technique	DSP Technique	Overall Accuracy	Advantages	Disadvantages	Dataset Type
[81]	Probabilistic Neural Network (PNN)	S-Transform	Pure signal – 99.26 % Signal with 40 dB noise – 99.13 % Signal with 30 dB noise – 98.63 %	By adapting the width factor in the S-transform, the method enhances the resolution of both time and frequency domains, which improves feature extraction accuracy.	robustness in different operational conditions. The approach's reliance on the precise adjustment of the width factor and feature selection might limit its generalizability across different types of PQ disturbances and operational conditions.	Synthetic
[82]	Deep Convolutional Neural Networks (DCNNs) and multiclass SVM	Curvelet Transform (CT)	100 %	Combining SSA and FDCT methods with DCNNs for feature extraction and classification enhances the system's accuracy and robustness.	Training DCNNs effectively requires substantial computational power and large datasets, which could be a limitation in environments with limited resources.	Synthetic
[83]	Fully Complex-valued radial basis function network (FC-RBF) and Complex extreme learning machine (CELM)	S-Transform	100 %	Compared to traditional real-valued classifiers like ELM and SVM, the complex-valued classifiers, especially FC-RBF, show better convergence and accuracy.	Training and tuning complex-valued networks can be more challenging and time-consuming, which may affect the practicality of deployment in resource-constrained environments.	Synthetic
[84]	Compressed sensing (CS) and Sparse autoencoder (SAE)	-	99–100 %	CS enables efficient feature extraction from PQD waveforms by sampling below the Nyquist rate, thus reducing the computational load. SAE further refines the feature set by removing redundant features, enhancing the overall classification performance.	The heavy dependence on synthetic data for validation may also limit the method's ability to generalize across diverse real-world scenarios.	Synthetic
[85]	Compressive sensing (CS), Singular spectrum analysis (SSA), and Deep neural network (DNN)	Wavelet Transform	99–100 %	CS efficiently reduces the dimensionality of features, aiding in faster and more effective classification.	The DNN-based approach and CS may demand substantial computational resources, which could be a constraint in resource-limited scenarios.	Synthetic
[87]	Hierarchical extreme learning machine (H-ELM)	Discrete Wavelet Transform	93.0 %	The hierarchical extreme learning machine (H-ELM) classifier accelerates the training process and provides robust classification without the need for extensive parameter tuning.	H-ELM, despite its faster training speed, lacks the flexibility of deep learning models in terms of handling complex nonlinear relationships between features.	Synthetic
[89]	Deep Convolutional Neural Networks	Fourier Transform	99.67 %	The transformation of 1D voltage disturbance signals into 2D images via WVD simplifies complex signal processing problems, turning them into image classification tasks that CNNs handle efficiently.	The reliance on WVD for 2D transformation may introduce artifacts or distortions in the signal, potentially affecting classification accuracy, particularly for subtle or overlapping disturbances.	Synthetic
[90]	Multi-Objective Grey Wolf Optimizer (MOGWO) with KNN	Two-dimensional Riesz Transform (RT)	99.26 %	MOGWO-KNN is used to select the most optimal features, creating a simpler, high-performance classification model.	The requirement for a large number of features (180 features for each sub-band) can make the model data-intensive, potentially leading to overfitting, especially with smaller datasets.	Synthetic
[91]	Deep Neural Network (DNN)	Hyperbolic window S-Transform	99.86 %	The proposed framework effectively automates feature extraction from the Hyperbolic Stockwell Transform (HWST) time-frequency spectrum using a Stacked Autoencoder (SAE).	The complexity of the framework, involving HWST, SAE, and XGBoost, increases computational demands, making real-time implementation potentially challenging, especially for large-scale power systems with real-time monitoring requirements.	Synthetic & Real
[92]	Deep Convolutional Neural Network (DCNN) and multiclass SVM	Segmented and Modified S-Transform	Pure signal – 99.54 % Signal with 20 dB noise – 98.86 %	The hybrid approach combining SMST, DCNN, and MSVM offers superior time-frequency resolution and accurate feature	A limitation of the method is the manual determination of the three adjustable parameters in the SMST, which can hinder optimal	Synthetic

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Table 6 (continued)

Ref.	AI Technique	DSP Technique	Overall Accuracy	Advantages	Disadvantages	Dataset Type
			Signal with 30 dB noise – 99.52 % Signal with 40 dB noise – 99.6 %	extraction for power quality (PQ) disturbances.	time-frequency resolution and classification accuracy.	
[93]	Compressed Neural Network Methods	Discrete Orthogonal S-Transforms	99.81 %	DOST ensures superior time-frequency localization, particularly at low and high frequencies, while the CS method reduces computational burden by compressing the feature matrix.	A key limitation of the method is its reliance on a complex pipeline of DOST, CS, and DSN, which may require careful tuning and expertise to implement.	Synthetic
[94]	Improved Grey Wolf Optimization-Based Kernel Extreme Learning Machine	Stock Well Transform	Signal with 40 dB noise – 98.86 % Signal with 30 dB noise – 98.32 % Signal with 20 dB noise – 97.3 %	IGWO enhances KELM's classification accuracy by optimizing the model parameters, making it an effective meta-heuristic method for fine-tuning.	The method involves tuning several parameters (ST, IGWO, KELM), which can be time-consuming and computationally expensive, especially for real-time applications.	Synthetic
[95]	GA-EML	Wavelet Transform, Hilbert Huang transform	99 %	The paper introduces a robust multi-dimensional feature extraction approach using AST, HHT, and WT, ensuring comprehensive PQD representation. Additionally, the hybrid feature selection and multi-classifier synergy significantly enhance classification accuracy	While effective, the method relies on complex feature extraction and selection, which may increase computational overhead.	Synthetic
[96]	1D- CNN	Discrete wavelet transform	-	The study proposes an enhanced 1D-CNN integrated with DWT-MRA for automatic PQD classification, reducing manual feature selection.	The reliance on simulated data limits real-world applicability, as grid conditions may vary significantly.	Synthetic
[97]	CNN	Wavelet Transform	99.8 %	The method simplifies PQD classification by leveraging CNNs' image-processing capabilities, removing manual feature extraction and selection while maintaining high accuracy.	The transformation of signals into images increases computational requirements, and the approach may struggle with real-time applications due to processing overhead.	Synthetic & Real
[98]	CNN	Stockwell Transform	>99 % in all cases	The method effectively captures PQD features using ST and optimizes CNN hyperparameters with BOA, leading to high classification accuracy.	The transformation process and hyperparameter optimization add computational complexity, making real-time implementation challenging.	Synthetic
[99]	ELM	Histogram of oriented gradients (HOG)	99.30 %	The HOG-ELM framework reduces computational complexity, making it faster and more efficient for real-time PQD detection.	The method's performance might degrade in highly noisy environments or complex real-world scenarios due to its reliance on predefined feature extraction techniques.	Synthetic & Real
[100]	LSTM	Wavelet Transform	With highest noise - 89.77 %	The attention-based LSTM model effectively captures both local and sequential dependencies, improving robustness against noise.	The computational complexity of wavelet decomposition and LSTM training may lead to increased processing time, making real-time implementation challenging.	Synthetic
[101]	Neural Networks	-	-	The use of adaptive linear networks (ADALINE) and feedforward neural networks (FFNN) provides robustness against noise, allowing for accurate detection and classification even in noisy environments.	While the dual-network approach effectively handles several types of disturbances, the method may require further enhancement to classify other complex or rarer disturbances not covered in this study.	Synthetic

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Table 6 (continued)

Ref.	AI Technique	DSP Technique	Overall Accuracy	Advantages	Disadvantages	Dataset Type
[102]	Variational Mode Decomposition and Deep Stochastic Configuration Network	-	99.4 %	The proposed approach has been tested with both analytical and actual measurements, ensuring that it is robust and reliable under different operational scenarios.	The paper suggests that future research is needed for online implementation. Without real-time application, the practicality of the method for continuous power system monitoring and control remains uncertain.	Synthetic
[103]	Deep Convolutional Neural Network (DCNN)	-	Signal without noise – 99.96 % Signal with 40 dB noise – 99.95 % Signal with 30 dB noise – 99.66 % @Signal with 20 dB noise – 98.13 %	The paper introduces a fully closed-loop deep convolutional neural network (CNN) approach that merges signal analysis, feature selection, and classification into one automatic process.	While CNN achieves high accuracy, its model size could be a concern for deployment on embedded devices or real-time systems.	Synthetic
[104]	Distribution Network	-	Signal with 30 dB noise – 99.38 % Signal with 20 dB noise – 96.75 % Signal with 10 dB noise – 94.82 %	The proposed algorithm improves classification accuracy by effectively detecting amplitude, frequency, and the start time of disturbances, while maintaining robustness across various power quality disturbance signals.	The paper lacks a design for an online real-time detection system, limiting its immediate practical application in real-time scenarios, especially in embedded systems.	Synthetic
[105]	Long Short Term Memory (LSTM) and Sliding Time Window	-	Signal with 50 dB noise – 99.91 % Signal with 30 dB noise – 99.88 % Signal with 20 dB noise – 99.64 %	The proposed method offers high accuracy in recognizing the type and locating the starting and ending times of power quality disturbances, thanks to the deep learning-based Long Short Term Memory (LSTM) model combined with a sliding time window.	The method may have higher computational complexity due to the transformation of disturbance waveforms into grayscale images and the use of deep learning, potentially requiring more computational resources compared to traditional methods.	Synthetic
[106]	K-means clustering	-	-	The proposed technique provides high flexibility and accuracy in identifying PQDs such as swells, sags, and interruptions.	Unsupervised learning techniques like K-means clustering may have difficulty accurately classifying events in complex or highly variable datasets, as they rely on predefined centroids, which may not always align perfectly with real-world disturbances.	Synthetic
[107]	1D and 2D Convolutional Neural Networks	-	99.97 %	The hybrid CNN method effectively leverages both 1D and 2D feature extraction by processing signals in their original form (1D CNN) and as images (2D CNN). This dual approach mimics how a power quality disturbance (PQD) expert would evaluate signals, enhancing classification performance by capturing a broader range of features.	While the method achieves high classification accuracy, its computational complexity remains like other advanced methods. This might pose a challenge in real-time applications, especially in systems with limited computational resources.	Synthetic & Real
[109]	Convolutional Neural Network (CNN)	-	Signal with 50 dB noise – 98.27 % Signal with 40 dB noise – 98.45 % Signal with 30 dB noise – 98.52 %	The proposed deep convolutional network, incorporating a modified Inception-ResNet structure, offers high classification, faster convergence, and strong generalization ability, making it a robust solution for PQDs	The method cannot detect certain disturbance signal components, such as amplitude, limiting its effectiveness in capturing all aspects of power quality disturbances. Further improvements are needed to address these limitations and	Synthetic

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Table 6 (continued)

Ref.	AI Technique	DSP Technique	Overall Accuracy	Advantages	Disadvantages	Dataset Type
[110]	Single-shot PQD detection (SSPQDD)	-	96.55 %	identification and classification in microgrids. The proposed Single Shot Power Quality Disturbance Detection (SSPQDD) method demonstrates high accuracy (96.55 %) and efficiency in detecting and classifying multiple disturbances within a single window frame.	extend its applicability to more power quality fields. While effective in most cases, the simulated nature of the dataset used for training and validation may limit its performance in real-world scenarios where disturbances can be more complex and varied.	Synthetic
[111]	Convolutional Neural Network (CNN) (Attention Model)	-	99.92 %	The attention model enhances the classification by effectively processing and rescaling the data, which leads to high performance in identifying various power quality conditions.	The method relies on large and potentially complex datasets with high dimensionality (622 data points and 5600 parameters), which might require significant computational resources and time for training.	Synthetic
[112]	Combined CNN and LSTM model	-	99.63 %	The approach benefits from automatic feature extraction and selection, reducing the reliance on conventional methods and improving performance, especially in noisy environments.	The method has been validated only in offline environments, and its performance in real-time or online scenarios remains untested.	Synthetic
[113]	Long Short-Term Memory (LSTM)	-	92.68 %	By utilizing synthetic voltage sag signals generated from mathematical formulas, the approach eliminates manual feature extraction and speeds up data preparation.	The approach also requires a significant amount of labeled synthetic data, which, although useful, may not always perfectly represent actual conditions.	Synthetic
[114]	Deep Auto-encoder Networks (DAE)	-	99.4 %	By avoiding complex signal processing techniques and using a simple classifier like SoftMax, the method significantly reduces computation time and complexity.	The primary limitation of this approach is the reliance on a smaller dataset, which, while effective in this study, may impact the generalizability of the model to different or larger datasets.	Synthetic
[115]	Transformer based model	-	91.43 % - 99.81 %	The proposed framework achieves high accuracy in classifying PQEs under challenging conditions without requiring separate feature extraction.	The framework's performance slightly decreases with high noise levels or extreme variations in signal amplitude and frequency.	Synthetic
[116]	Vision Transformer	-	90.94 % - 99.73 %	The proposed ViT-SPWVD method achieves superior PQD classification accuracy by leveraging self-attention mechanisms and converting 1-D signals to 2-D images.	The technique relies on synthetic PQD data generation, which may not fully capture real-world complexities or generalize effectively to noisy data.	Synthetic

achieving the purpose of preprocessing, Dual-Tree Complex Wavelet Transform (DTCWT), a DSP technique was utilized. Due to the lack of shift-invariance and poor directionality in the conventional technique, Discrete Wavelet Transform (DWT), DTCWT, was developed. It is capable of accurately extracting features from power system transient signals and providing better results [123]. It yielded complex matrices, which were further plotted as transformed curves, namely magnitude and phase plots as shown in Fig. 8. From these curves, ten features were extracted, including energy, standard deviation, mean, skewness, and kurtosis of both plots. The statistical data from these features was then used as an input for the AI classifier to execute classification. The Classification Learner app in MATLAB was employed to identify PQ disturbances. For each disturbance type, 20 sets of data were generated across different parameter ranges to evaluate the classifier's accuracy. The dataset was split, with 70 % used for training and 30 % for testing. Specifically, for each disturbance, 14 sets were allocated for training and 6 sets for testing.

The Decision Tree classifier was tested with three configurations using the Gini diversity index as the split criterion. The Fine Tree allowed a maximum of 100 splits for detailed decision boundaries, the Medium Tree used 20 splits for a balanced complexity, and the Coarse Tree limited splits to 4 for simplicity. The Discriminant Analysis classifier was evaluated with two configurations: Linear Discriminant Analysis and Quadratic Discriminant Analysis, both using a full covariance structure to model the data. The Support Vector Machine (SVM) classifier was configured with various kernels. Linear, Cubic, and Quadratic SVMs used an automatic kernel scale, a box constraint of 1, and the one-vs-one multiclass method. Medium, Coarse, and Fine Gaussian SVMs employed kernel scales of 3.2, 13, and 0.79, respectively, while maintaining the same box constraint and multiclass method. The Nearest Neighbour classifier

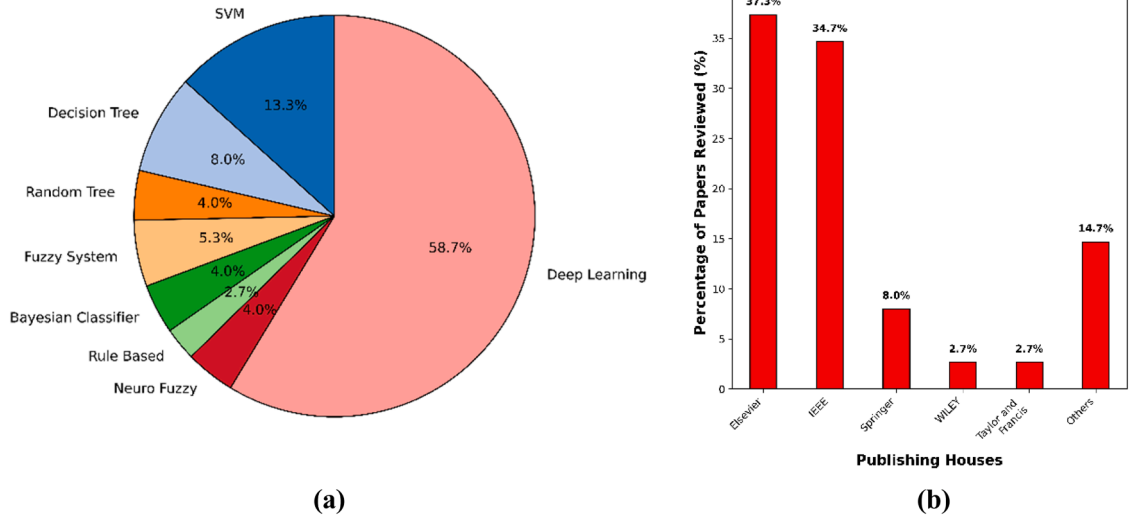


Fig. 7. (a) Distribution of Paper Reviewed by ML/DL technique (b) Papers Reviewed by the Authors in this Manuscript.

Table 7

Synthetic model and parameters of PQDs.

PQ disturbance	Mathematical model	Parameter range
Sag	$p(t) = (1 - \theta(u(t - t^{\#}) - u(t - t^{\#\#})))\sin\omega t$	$0.1 \leq \theta \leq 0.9, T \leq t^{\#\#} - t^{\#} \leq 9T$
Flicker	$p(t) = (1 + \theta_f \sin(\beta\omega_f t))\sin\omega t$	$0.1 \leq \theta_f \leq 0.2, 5 \leq \beta \leq 20\text{Hz}$
Interruption	$p(t) = (1 - \theta(u(t - t^{\#}) - u(t - t^{\#\#})))\sin\omega t$	$0.9 \leq \theta \leq 1.0, T \leq t^{\#\#} - t^{\#} \leq 9T$
Swell	$p(t) = (1 + \theta(u(t - t^{\#}) - u(t - t^{\#\#})))\sin\omega t$	$0.1 \leq \theta \leq 0.8, T \leq t^{\#\#} - t^{\#} \leq 9T$
Oscillatory transient	$p(t) = \sin\omega t + \theta \exp\left(-\frac{(t - t^{\#})}{\tau}\right)\sin\omega_f(t - t^{\#})\{u(t^{\#\#}) - u(t^{\#})\}$	$0.1 \leq \theta \leq 0.8, 0.5T \leq t^{\#\#} - t^{\#} \leq 3T, 8\text{ms} \leq \tau \leq 40\text{ms}, 300 \leq f_f \leq 900\text{Hz}$
Flicker with harmonics	$p(t) = (1 + \theta_f \sin(\beta\omega_f t))(\theta_1 \sin(\omega t) + \theta_3 \sin(3\omega t) + \theta_5 \sin(5\omega t) + \theta_7 \sin(7\omega t))$	$0.1 \leq \theta_f \leq 0.2, 5 \leq \beta \leq 20\text{Hz}, 0.05 \leq \theta_3, \theta_5, \theta_7 \leq 0.15, \sum \theta_i^2 = 1$
Sag with harmonics	$p(t) = (1 - \theta(u(t - t^{\#}) - u(t - t^{\#\#})))(\theta_1 \sin(\omega t) + \theta_3 \sin(3\omega t) + \theta_5 \sin(5\omega t) + \theta_7 \sin(7\omega t))$	$0.1 \leq \theta \leq 0.9, T \leq t^{\#\#} - t^{\#} \leq 9T, 0.05 \leq \theta_3, \theta_5, \theta_7 \leq 0.15, \sum \theta_i^2 = 1$
Transient with harmonics	$p(t) = (\theta_1 \sin(\omega t) + \theta_3 \sin(3\omega t) + \theta_5 \sin(5\omega t) + \theta_7 \sin(7\omega t)) + \theta \exp\left(-\frac{(t - t^{\#})}{\tau}\right)\sin\omega_f(t - t^{\#})\{u(t^{\#\#}) - u(t^{\#})\}$	$0.1 \leq \theta \leq 0.8, 0.5T \leq t^{\#\#} - t^{\#} \leq 3T, 8\text{ms} \leq \tau \leq 40\text{ms}, 300 \leq f_f \leq 900\text{Hz}$
Swell with harmonics	$p(t) = (1 + \theta(u(t - t^{\#}) - u(t - t^{\#\#})))(\theta_1 \sin(\omega t) + \theta_3 \sin(3\omega t) + \theta_5 \sin(5\omega t) + \theta_7 \sin(7\omega t))$	$0.1 \leq \theta \leq 0.8, T \leq t^{\#\#} - t^{\#} \leq 9T, 0.05 \leq \theta_3, \theta_5, \theta_7 \leq 0.15, \sum \theta_i^2 = 1$
Interruption with harmonics	$p(t) = (1 - \theta(u(t - t^{\#}) - u(t - t^{\#\#})))(\theta_1 \sin(\omega t) + \theta_3 \sin(3\omega t) + \theta_5 \sin(5\omega t) + \theta_7 \sin(7\omega t))$	$0.9 \leq \theta \leq 1.0, T \leq t^{\#\#} - t^{\#} \leq 9T, 0.05 \leq \theta_3, \theta_5, \theta_7 \leq 0.15, \sum \theta_i^2 = 1$

was tested with different configurations. Fine KNN used 1 neighbor with the Euclidean distance metric and equal distance weight. Medium and Coarse KNN used 10 and 100 neighbors, respectively, both with Euclidean distance and equal weighting. Cubic KNN used 10 neighbors with the Minkowski (Cubic) metric and equal weights. Weighted KNN also used 10 neighbors and Euclidean distance but applied squared inverse weighting. Lastly, Cosine KNN employed 10 neighbors with the Cosine distance metric and equal weights. The Ensemble Classifier was tested with various configurations. Boosted Trees and RUSBoosted Trees used Decision Trees with a maximum of 20 splits, 30 learners, and a learning rate of 0.1, with RUSBoost employing the RUS Boost method. Bagged Trees used Decision Trees with 139 splits and 30 learners. Subspace Discriminant and Subspace KNN used 30 learners, with the subspace dimension set to 5; Subspace KNN utilized the subspace ensemble method. The Neural Network Classifier was configured with different architectures. Narrow, Wide, and Medium Neural Networks used one fully connected layer with sizes of 10, 25, and 100 neurons, respectively, and ReLU activation. Trilayered and Bilayered Neural Networks had 2 and 3 layers with sizes of 10,10 and 10,10,10, respectively. All models had an iteration limit of 1000 and a regularization strength of 0.

After testing the classifier, as outlined in Table 8, the Bagged Trees classifier achieved the highest testing accuracy of 96.6 %. This was closely followed by Coarse Gaussian SVM and Fine KNN achieving 93.33 % and 91.67 % respectively. In Decision Trees, Fine and Medium Trees demonstrated high training accuracy (95 %) but relatively lower testing accuracy (75 %), indicating a possible issue of overfitting. Linear Discriminant under Discriminant Analysis performed well, achieving 96.4 % training and 91.67 % testing accuracy, while Quadratic Discriminant failed. Naïve Bayes classifiers showed perfect training accuracy (100 %), but testing accuracy declined slightly, with Kernel Naïve Bayes at 73.33 % and Gaussian Naïve Bayes at 70 %. Nearest Neighbor Classifiers had varied results, with Fine KNN and Cosine KNN achieving competitive testing accuracy of 91.67 % and 90 %, respectively. In Ensemble methods, the Subspace Discriminant achieved 95 % testing accuracy. Neural Networks displayed mixed results, with Narrow Neural Networks

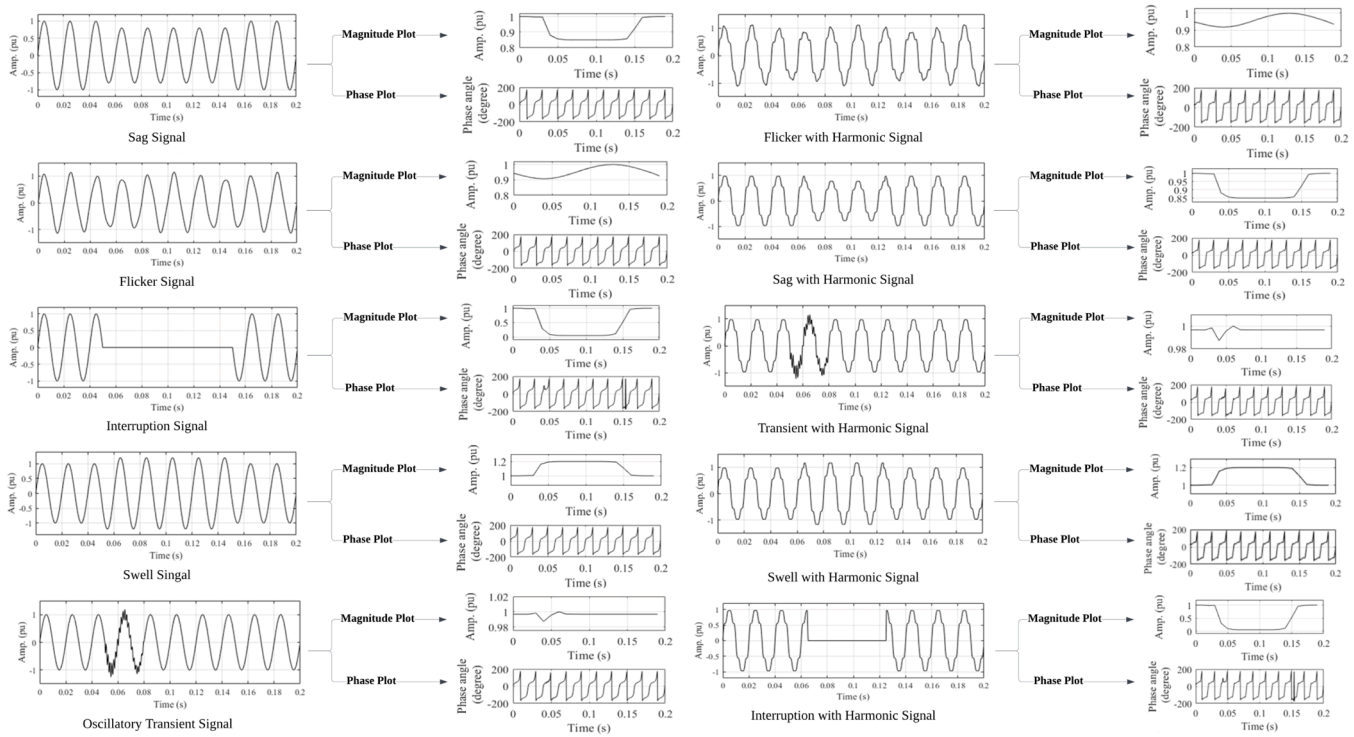


Fig. 8. Magnitude and Phase Plots for Various PQDs.

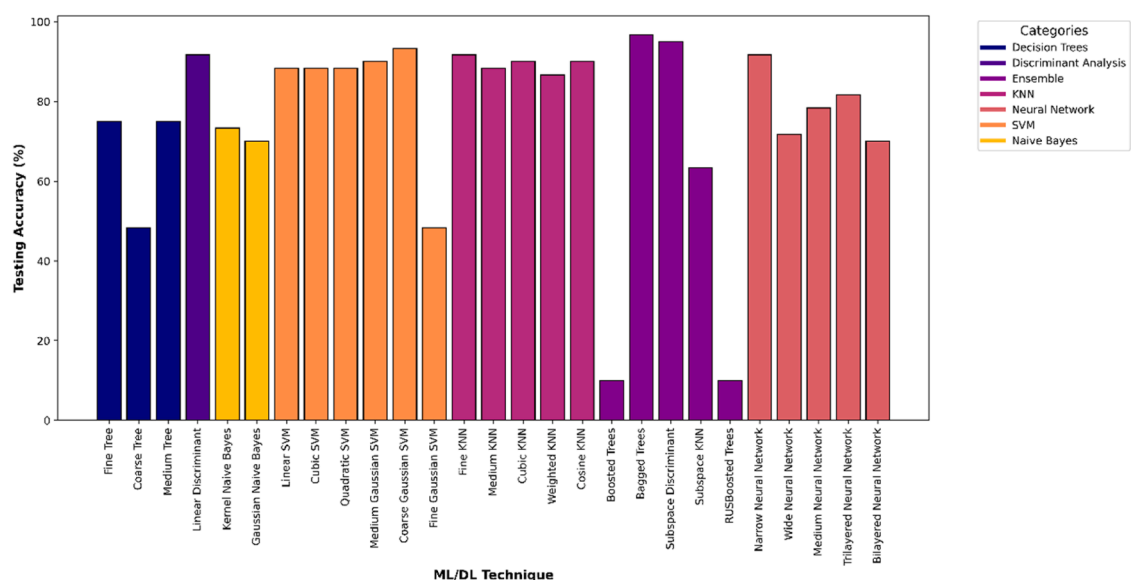
Table 8

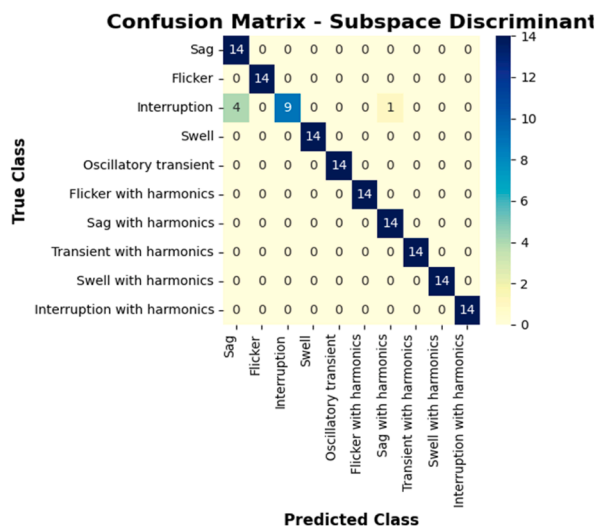
Comparison of training and testing accuracy across various ML/DL techniques for PQD classification.

ML/DL Technique	Sub Classification	Training Accuracy (%)	Testing Accuracy(%)
Decision Trees	Fine Tree	95	75
	Coarse Tree	44.3	48.33
	Medium Tree	95	75
Discriminant Analysis	Linear Discriminant	96.4	91.67
	Quadratic Discriminant	Failed	-
Naïve Bayes Classifier	Kernel Naïve Bayes	100	73.33
	Gaussian Naïve Bayes	100	70
Support Vector Machines	Linear SVM	97.9	88.33
	Cubic SVM	97.9	88.33
	Quadratic SVM	99.3	88.33
	Medium Gaussian SVM	99.3	90
	Coarse Gaussian SVM	98.6	93.33
	Fine Gaussian SVM	97.9	48.33
Nearest Neighbor Classifier	Fine KNN	98.6	91.67
	Coarse KNN	07.1	0
	Medium KNN	94.3	88.33
	Cubic KNN	92.9	90
	Weighted KNN	97.9	86.67
	Cosine KNN	93.6	90
Ensemble Classifier	Boosted Trees	07.1	10
	Bagged Trees	98.6	96.67
	Subspace Discriminant	96.4	95
	Subspace KNN	81.4	63.33
	RUSBoosted Trees	07.1	10
Neural Network Classifier	Narrow Neural Network	96.4	91.67
	Wide Neural Network	95.7	71.67
	Medium Neural Network	94.3	78.33
	Trilayered Neural Network	90	81.67
	Bilayered Neural Network	96.4	70

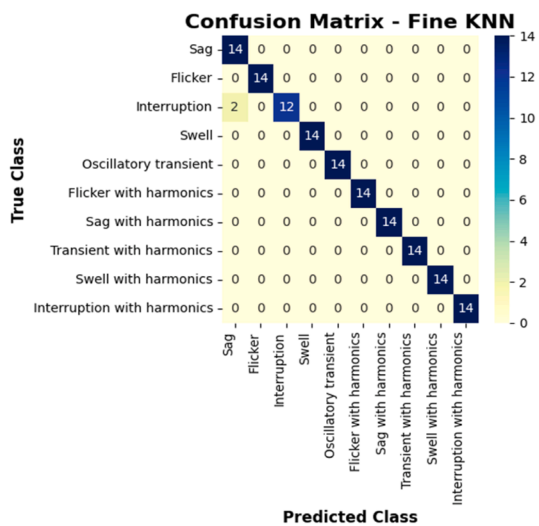
reaching a notable testing accuracy of 91.67 %, while other configurations showed moderate performance. The performance of the top 6 classifiers is illustrated in the confusion matrix, as shown in Fig. 10. Fig. 9 presents a comparative analysis of various ML and DL techniques, illustrating the testing accuracies of each.

The exceptional results underscore the potential of these classifiers to address critical challenges faced by modern power systems. Voltage sags, swells, transients, and harmonics can severely affect devices like medical equipment and industrial machinery, leading to malfunctions and increased maintenance costs. Accurate and reliable PQD detection mitigates the adverse impacts of power quality issues, reducing maintenance costs and extending the lifespan of infrastructure across residential, commercial, and industrial applications. This robust framework also facilitates the seamless integration of renewable energy sources into the grid, supporting global

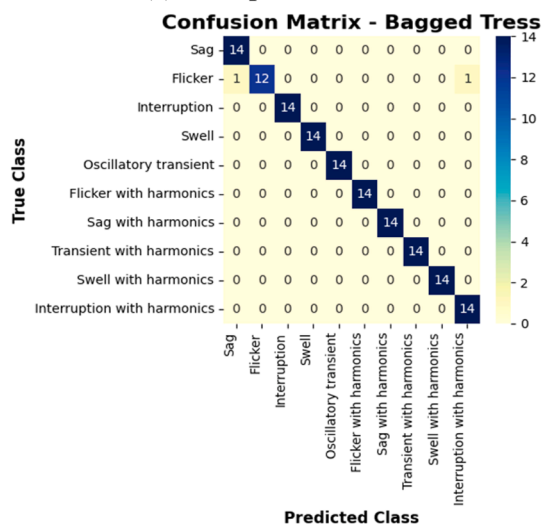
**Fig. 9.** Performance of ML and DL Techniques for PQD Classification.



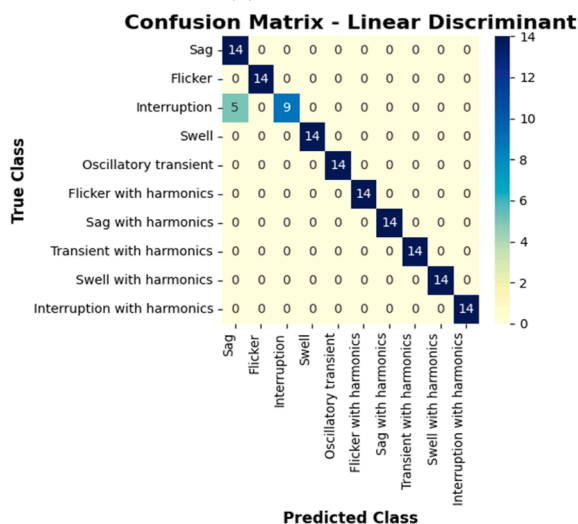
(a) Subspace Discriminant



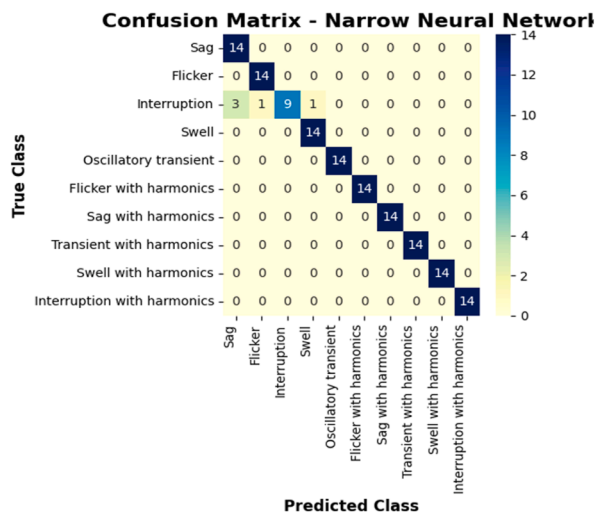
(b) Fine KNN



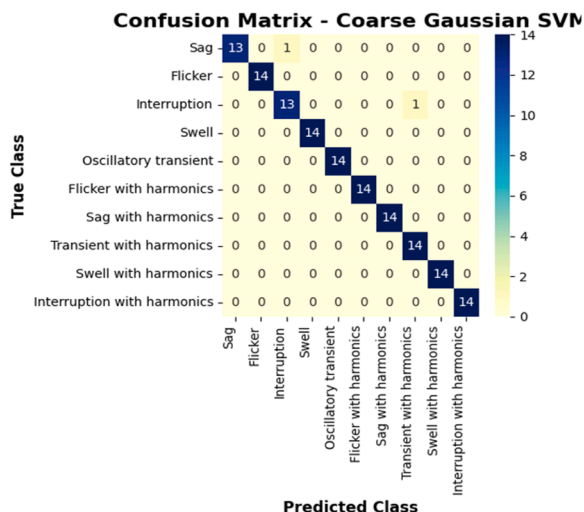
(c) Bagged Trees



(d) Linear Discriminant



(e) Narrow Neural Network



(f) Coarse Gaussian SVM

Fig. 10. Confusion Matrix of Top 6 Classifiers.

efforts toward sustainable energy solutions. It ultimately benefits society by ensuring a stable, efficient, and environmentally responsible power supply. Governments can leverage these advanced PQD detection methods to develop policies that promote the adoption of smart grid technologies, implement stricter regulations on power quality standards, and incentivize industries to adopt better power management practices.

7. Conclusion and future scope

With the global shift toward clean energy and smart grids, the detection of PQDs becomes increasingly vital, as these disruptions can severely damage grid infrastructure, leading to inefficiency, equipment failure, and malfunctioning industrial processes. Numerous advancements have been made in PQD detection, with Artificial Intelligence (AI) techniques playing a pivotal role in significantly enhancing the accuracy and efficiency of these systems. The integration of AI has brought about remarkable improvements in data analysis, pattern recognition, and real-time monitoring. This paper presents a thorough examination of AI methods for PQD classification as well as approaches incorporating DSP and non-DSP techniques. These AI methods encompass SVM, Decision Trees, Rule-Based Classifiers, Fuzzy Logic, Bayesian Classifiers, and Deep Learning methods. The reviewed papers demonstrate the substantial benefits of AI in this field, showcasing its transformative impact.

A case study was conducted to evaluate the performance of various machine learning classifiers in classifying PQDs, extending the analysis of reviewed AI techniques. Among the tested classifiers, Bagged Trees achieved the highest testing accuracy of 96.6 %. By reviewing high-quality studies addressing various PQDs, the paper highlights their methodologies, advantages, and limitations, offering a detailed comparative analysis. Additionally, the explanation of the algorithms of each AI technique fosters a deeper understanding of their application to PQD classification. Not only does this paper play a crucial role in shaping future studies, but it also paves the way for a deeper understanding of more reliable power grids, which are crucial for sustainable energy management. This aligns with Sustainable Development Goal 7 which emphasizes affordable, reliable, and sustainable energy for all. Ultimately, the study inspires technological innovation and promotes societal well-being by ensuring uninterrupted power supply and supporting the transition to sustainable energy sources.

Despite notable progress, certain limitations persist in the field. Research on state-of-the-art deep learning models for PQD detection remains scarce, indicating a need for further exploration. Additionally, handling complex PQDs with overlapping or non-standard patterns poses considerable challenges. Furthermore, the inadequate focus on identifying the underlying causes of PQDs, is essential for implementing effective mitigation strategies. Despite these issues, the advancements achieved through AI represent a significant leap forward.

CRedit authorship contribution statement

Shaurya Jain: Methodology, Software, Writing – original draft, Data curation. **Amol Satsangi:** Methodology, Visualization, Software. **Rajat Kumar:** Conceptualization, Supervision, Validation. **Divyani Panwar:** Resources, Project administration. **Mohammad Amir:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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